Quantum Computing - Notes - v0.4.0

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Preface

Every theory section in these notes has been taken from the sources:

- Course slides. [1]

About:

GitHub repository

These notes are an unofficial resource and shouldn't replace the course material or any other book on quantum computing. It is not made for commercial purposes. I've made the following notes to help me improve my knowledge and maybe it can be helpful for everyone.

As I have highlighted, a student should choose the teacher's material or a book on the topic. These notes can only be a helpful material.

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1 Introduction

1.1 Complex Numbers

Let's not start from math. Let's start from **physics / computation intuition**. Classical bits are represented by two distinct states: 0 and 1. In quantum computing, we have **qubits**, which are essentially **vector of numbers**. But not just any numbers: a qubit is a vector of **complex numbers**. They are needed because quantum systems have **both magnitude and phase**, and complex numbers can represent both. Real numbers can only represent "*how much*" (magnitude), while complex numbers can represent "*how much*" and "*in which direction / phase*" (magnitude and phase).

Why complex (and not just real) numbers?

1. In quantum computing, we don't store probabilities directly (e.g., 0.5 for a 50% chance). Instead, we store **probability amplitudes**, which are complex numbers. The probability of an outcome is the square of the magnitude of its amplitude:
$$P = |z|^2$$

2. **Interference**. Quantum states can **combine and cancel**. For example, if we have two states with amplitudes z_1 and z_2, the combined amplitude is:
$$z_{\text{combined}} = z_1 + z_2$$
If z_1 and z_2 have the same magnitude (*how much*) but opposite phases (*in which direction*), they can cancel each other out, leading to zero probability for that outcome. This is a key feature of quantum mechanics that allows for phenomena like quantum interference, which is essential for quantum algorithms.

In general, complex numbers are needed because **quantum states behave like waves**, not like classical values. Waves have both amplitude and phase, and complex numbers are the natural way to represent them mathematically. This is why quantum mechanics and quantum computing rely heavily on complex numbers.

What do we actually need?

Not everything. Just a **minimal toolkit**. A complex number is:

$$z = x + iy$$

Where x is the **real part** and y is the **imaginary part**, and i is the **imaginary unit** satisfying $i^2 = -1$. From a geometric interpretation, we can think of z as a point in the 2D plane, where x is the horizontal coordinate (real axis) and y is the vertical coordinate (imaginary axis).

Polar form

Instead of coordinates (x, y), we can also represent a complex number in **polar form**:

$$z = re^{i\varphi}$$

Where:

- $r = |z| = \sqrt{x^2 + y^2}$ is the **modulus** (or *magnitude*) of the complex number, representing its distance from the origin in the complex plane.
- $\varphi = \arg(z) = \tan^{-1}\left(\frac{y}{x}\right)$ is the **argument** (or *phase* angle), representing the angle between the positive real axis and the line connecting the origin to the point z in the complex plane.
- The **exponential form** $e^{i\varphi}$ means that we can represent the complex number as a point on the unit circle in the complex plane, where φ is the angle from the positive real axis. The modulus r then scales this point to its actual distance from the origin.

  Using Euler's formula:

  $$e^{i\varphi} = \cos\varphi + i\sin\varphi$$

  We can rewrite z as:

  $$z = r\left(\cos\varphi + i\sin\varphi\right)$$

  This form is particularly useful in quantum computing because it makes it easy to understand how complex numbers can represent both magnitude and phase, which are essential for describing quantum states and their evolution.

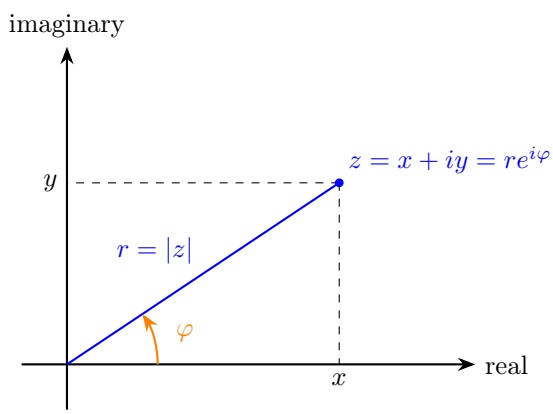


Figure 1: Geometric interpretation of a complex number in the complex plane. The arrow represents the complex number z as a vector from the origin to the point (x, y). The length of the vector is the modulus r, and the angle it makes with the positive real axis is the argument φ.

Complex Conjugate

The **complex conjugate** of a complex number $z = x + iy$ is denoted as $\bar{z}$ and is defined as:

$$\bar{z} = x - iy$$

Geometrically, the complex conjugate $\bar{z}$ is the **reflection** of z **across the real axis in the complex plane**. This means that if z is represented by the point (x, y), then $\bar{z}$ is represented by the point $(x, -y)$. The real part remains the same, while the imaginary part changes sign. See Figure 2 for a visual representation of this concept.

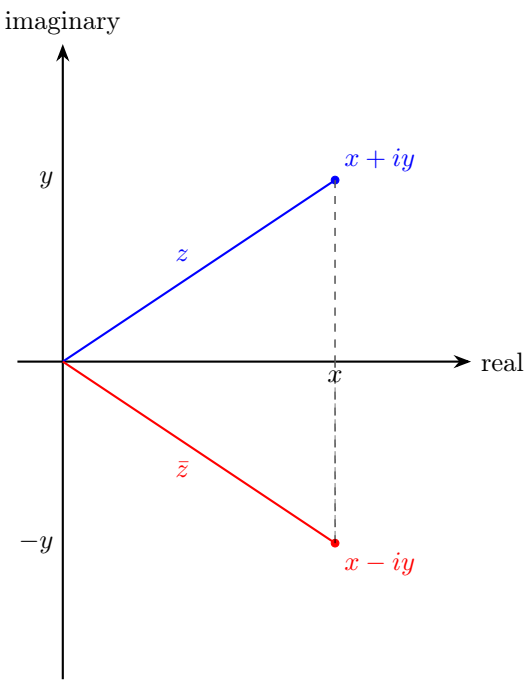


Figure 2: Geometric interpretation of the complex conjugate: reflection across the real axis.

Magnitude

The **magnitude** (or **modulus**) of a complex number $z = x + iy$ is denoted as $|z|$ and is defined as:

$$|z| = \sqrt{x^2 + y^2}$$

And it can also be expressed using the complex conjugate:

$$|z|^2 = z \cdot \bar{z} = (x + iy)(x - iy) = x^2 + y^2$$

The magnitude represents the **distance** of the point z from the origin in the complex plane. It is always a non-negative real number. The magnitude is important in quantum mechanics because the probability of an outcome is given by the square of the magnitude of the corresponding probability amplitude. For

example, if a quantum state has an amplitude of z, the probability of measuring that state is $|z|^2$. This is why the magnitude of complex numbers is crucial in quantum computing, as it directly relates to the probabilities of different outcomes when measuring quantum states.

🔄 Rotation in the complex plane

Multiplying a complex number by a complex exponential of the form $e^{i\theta}$ corresponds to a **rotation** in the complex plane:

$$z' = z \cdot e^{i\theta} = re^{i(\varphi+\theta)}$$

Geometrically, this operation:

- **Preserves the magnitude** $|z|$
- **Adds** θ **to the phase** φ

Therefore, multiplication by $e^{i\theta}$ rotates the complex number by an angle θ around the origin.

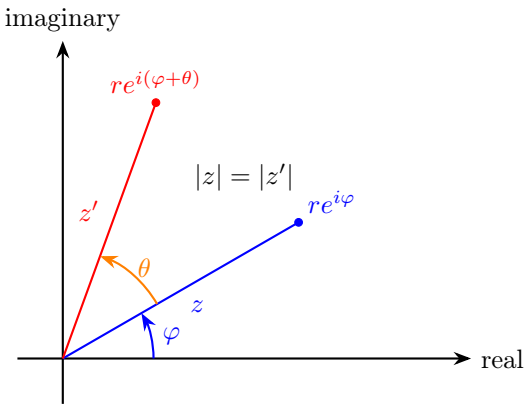


Figure 3: Rotation of a complex number: multiplying by $e^{i\theta}$ rotates the vector by angle θ while preserving its magnitude.

⇄ Hermitian (Conjugate Transpose)

Given a complex vector:

$$\mathbf{v} = \begin{bmatrix} a \\ b \end{bmatrix}$$

Its **Hermitian** (or **conjugate transpose**) is defined as:

$$\mathbf{v}^\dagger = \begin{bmatrix} \bar{a} & \bar{b} \end{bmatrix}$$

This operation consists of:

- Taking the **transpose** (column → row)
- Taking the **complex conjugate** of each element

1.2 Dirac's Notation

Dirac notation was introduced to make **linear algebra with complex vectors easier, cleaner, and more expressive**, especially for quantum systems.

Let's imagine our work only with standard linear algebra notation, where we have to write out all the vectors and matrices in their full form:

- Vectors:
$$\mathbf{v} = \begin{bmatrix} a \\ b \end{bmatrix}$$
- Conjugate transpose:
$$\mathbf{v}^\dagger = \begin{bmatrix} \bar{a} & \bar{b} \end{bmatrix}$$
- Inner product:
$$\mathbf{v}^\dagger \mathbf{w} = \bar{a}c + \bar{b}d$$
- Matrix-vector multiplication:
$$M\mathbf{v} = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} m_{11}a + m_{12}b \\ m_{21}a + m_{22}b \end{bmatrix}$$

This quickly becomes **cumbersome and hard to read**, especially as we deal with more complex quantum systems. In quantum mechanics we constantly write expressions involving inner products, matrix operations, and state transformations.

Dirac's idea. Paul Dirac said "*let's give different roles* ***different symbols***". So he introduced:

- **Kets** $|v\rangle$ to represent **column vectors** (quantum states).
- **Bras** $\langle v|$ to represent row vectors (**conjugate transposes**).

Where the operators stay in the middle, and we can write everything in a much cleaner way:

- Vectors (**kets**):
$$|v\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$$
  Where the entries are complex numbers and usually we require $|v\rangle$ to have unit norm (normalized state).
- Conjugate transpose (**bra**):
$$\langle v| = \begin{bmatrix} \bar{a} & \bar{b} \end{bmatrix}$$
- Inner product:
$$\langle v|w\rangle = \bar{a}c + \bar{b}d$$
  The result is a scalar (complex number) that tells us how much the two states overlap.

- Matrix-vector multiplication:

$$M|v\rangle = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} m_{11}a + m_{12}b \\ m_{21}a + m_{22}b \end{bmatrix}$$

  An operator M acting on a state $|v\rangle$ gives us a new state. The result is a new ket.

Even if we ignore physics for now, we can see that Dirac's notation is more expressive and easier to read than standard linear algebra notation.

What is *Chaining* in Dirac notation?

Chaining means writing multiple operations together in a single expression without needing to write out intermediate steps. For example:

$$\langle x| M |y\rangle$$

It is just a **compact way to combine linear algebra operations** (matrix multiplication and inner products) in a single expression. It allows us to write complex quantum mechanical expressions in a much cleaner and more intuitive way.

Without Dirac notation, we would have to write:

$$\mathbf{x}^\dagger M \mathbf{y}$$

That is a **chain of operations**: take the conjugate transpose of $\mathbf{x}$, multiply it by M, and then multiply the result by $\mathbf{y}$. With Dirac notation, we can write this in a much more compact and readable way as $\langle x| M |y\rangle$.

To read $\langle x| M |y\rangle$, we can think of it as:

1. Start with the ket $|y\rangle$ (a quantum state).
2. Apply the operator M to $|y\rangle$, which gives us a new state $M|y\rangle$.
3. Take the inner product of $\langle x|$ with the new state $M|y\rangle$, which gives us a scalar (complex number) that represents the probability amplitude of transitioning from state $|y\rangle$ to state $|x\rangle$ via the operator M. At first, this might seem abstract, but as we go through more examples and applications in quantum mechanics and quantum computing, it will become much clearer how powerful and intuitive Dirac's notation is for describing quantum systems.

2 Single Qubit States

Before introducing qubits, we briefly describe what we mean by a **quantum system**.

A **Quantum System** is a physical system (such as a photon, an electron, or an atom) whose behavior is described by the laws of quantum mechanics.

In this course, we will focus on the simplest case: systems that can produce **two possible outcomes** when measured. These are called **two-level systems**, and they form the basis of qubits.

2.1 Building and Measuring Qubits (Intuition)

Before introducing the mathematical formalism of qubits, it is useful to develop an **intuitive understanding** of how a simple quantum system behaves when it is **prepared** and **measured**.

A Simple Quantum Experiment

Consider a very simple experimental setup:

- A **source** emits a quantum particle (for example, a photon).
- The particle passes through a device.
- A **detector** measures the outcome.

Each time we perform the experiment, the detector returns one of two possible results:

$$0 \quad \text{or} \quad 1$$

This motivates the idea of a **two-level system**, which is the simplest type of quantum system.

Key Observation

If we repeat the same experiment many times under identical conditions, we observe that:

- The outcome is **not deterministic**.
- Instead, it is **probabilistic**.

For example, we might obtain:

- Outcome 0 with probability p
- Outcome 1 with probability $1 - p$

This is fundamentally different from classical systems, where the outcome is determined in advance.

What is a Measurement?

A **Measurement** is the process that:

- Extracts **classical information** from a quantum system;
- Produces a **definite outcome** (e.g., 0 or 1).

An important property of quantum measurement is that it **affects the system itself**. After a measurement is performed, the **system is no longer in its original configuration**.

From Intuition to Formalization

This simple experiment suggests that:

- A quantum system can produce **two possible outcomes**
- The outcomes are governed by **probabilities**
- The system must therefore contain some form of **internal state** that determines these probabilities

In the following example, we will introduce a concrete physical example (photon polarization) that allows us to represent this internal state using **vectors with complex coefficients**.

Example 1: Example of a Two-Level Quantum System

To make the previous discussion more concrete, we introduce a simple physical system that behaves as a two-level quantum system: the **polarization of a photon**.

What is a Photon?

A **Photon** is the elementary particle of light. It carries energy and behaves both as a particle and as a wave. In this course, we will use photons as a simple example of a **quantum system**, since their properties can be easily controlled and measured.

What is Polarization?

Polarization describes the direction in which the electric field of a light wave oscillates. In simple terms, it can be thought of as the **orientation** of the light wave. For a photon, polarization can be visualized as a direction in space. In particular, we consider two basic polarization directions:

- **horizontal** ($\rightarrow$)
- **vertical** ($\uparrow$)

These two directions form a **two-level system**, since a measurement can distinguish between them (i.e., we can determine whether the photon is horizontally or vertically polarized).

Superposition of Polarizations

A photon is not restricted to being only horizontal or vertical. In general, it can be in a **combination** of both states:

$$v = a \rightarrow + b \uparrow$$

Where:

- a and b are **complex coefficients**;
- They determine the behavior of the system when measured.

This is the first example of a **quantum state**: a system described by a linear combination of basic states.

Vector Representation

We can represent these polarization states using vectors:

$$\rightarrow = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \uparrow = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

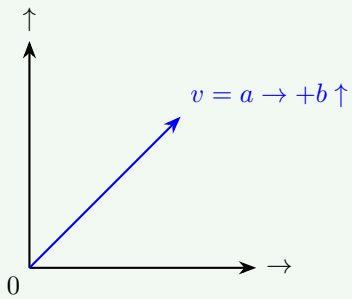
Therefore, a general polarization state becomes:

$$v = \begin{bmatrix} a \\ b \end{bmatrix}$$

This shows that the state of the system can be described as a **vector with complex components**, exactly as introduced in Dirac notation.

Geometric Interpretation

The horizontal and vertical polarization states can be interpreted as two **orthogonal axes** in a two-dimensional space.



The point where the axes intersect represents the origin, and any polarization state can be represented as a **vector starting from the origin**.

In this representation:

- $\rightarrow$ and $\uparrow$ form a **basis**
- Any state is a **linear combination** of these basis states

Measurement with Polarizers

A **Polarizer** is a device that **filters light based on its polarization**. A polarizer acts like a **filter** that allows only one polarization direction to pass and blocks the other. For example:

- A polarizer aligned with the horizontal direction ($\rightarrow$) will allow horizontally polarized light to pass and block vertically polarized light.
- A polarizer aligned with the vertical direction ($\uparrow$) will allow vertically polarized light to pass and block horizontally polarized light.

If the photon is a combination:

$$v = a \rightarrow + b \uparrow$$

And we use a:

- Horizontal polarizer ($\rightarrow$), the photon will pass with probability $|a|^2$ and be absorbed with probability $|b|^2$.
- Vertical polarizer ($\uparrow$), the photon will pass with probability $|b|^2$ and be absorbed with probability $|a|^2$.

This shows that:

- Probabilities are determined by the amplitudes (i.e., the coefficients a and b).
- Measurement **modifies the state of the system** (e.g., if the photon passes through the horizontal polarizer, it will be in the state $\rightarrow$).

2.2 From Polarization to Qubits

The example of photon polarization (page 12) shows that a physical system can be described using two basic states and their linear combinations. We now **abstract** this idea and introduce a **general model that does not depend on a specific physical system**.

From Physical States to Abstract States

In the polarization example, we used:

$$\rightarrow \quad \text{and} \quad \uparrow$$

As the two basic states of the system. We now replace them with an abstract notation:

$$\rightarrow \rightsquigarrow \ |0\rangle \qquad \uparrow \rightsquigarrow \ |1\rangle$$

These are called the **computational basis states**. They are just symbols (used in Dirac notation) that represent the two basic states of the system, without referring to any specific physical implementation.

Definition of a Qubit

A **Qubit** is a **two-level quantum system** whose state can be written as:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

It is the **unit of quantum information**, analogous to a classical bit. However, unlike a classical bit that can only be in one of two states (0 or 1), a qubit can exist in **infinitely many possible states**. Indeed, any normalized vector of the form:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

Represents a valid state of a qubit. Some **properties** of the qubit state are:

- The coefficients a and b are **complex numbers** (probability amplitudes). They determine the outcome of a measurement:
  - $|a|^2$ is the probability of measuring the state $|0\rangle$.
  - $|b|^2$ is the probability of measuring the state $|1\rangle$.

  Therefore, even though a qubit can be in infinitely many possible states, a measurement (in this basis) can only produce **two possible outcomes**: $|0\rangle$ or $|1\rangle$.

- The state is **normalized**:

  $$|a|^2 + |b|^2 = 1$$

  Because the probabilities must sum to 1 (i.e., we must get either $|0\rangle$ or $|1\rangle$ when we measure).

- The states $|0\rangle$ and $|1\rangle$ form a **basis** of the vector space describing the qubit.

  What is a *basis*? A **Basis** is a set of vectors that allows us to **represent any vector in the space as a linear combination of them**. In quantum computing, $|0\rangle$ and $|1\rangle$ are the **basis vectors** and any qubit state is built from them.

  In particular, the states $|0\rangle$ and $|1\rangle$ form an **Orthonormal Basis**, meaning that:

  - They are **orthogonal**: $\langle 0|1\rangle = 0$.
    Why is orthogonality important? Orthogonality means that the two basis states are completely distinguishable by a measurement (not overlapping). If we measure a qubit in the state $|0\rangle$, we will never get $|1\rangle$, and vice versa.
  - They are **normalized**: $\langle 0|0\rangle = \langle 1|1\rangle = 1$.
    Why is normalization important? Normalization ensures that the probabilities of measurement outcomes are well-defined and sum to 1. It also means that the basis states have unit length in the vector space, which is a fundamental property for quantum states.

Why This Abstraction Matters

This abstraction is fundamental because it allows us to:

- Describe **any** two-level quantum system (not only photons);
- Use a unified mathematical framework based on vectors;
- Apply the tools of linear algebra to quantum systems.

From now on, we will work with qubits using Dirac notation, without referring to a specific physical implementation.

Example 2: Different Choices of Basis

The computational basis $\{|0\rangle, |1\rangle\}$ is not the only possible choice. A qubit can be described using **any orthonormal basis**.

For example, another important basis is the **Hadamard basis**, defined as:

$$|+\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} \qquad |-\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$$

This shows that the same quantum state can be represented in different bases, depending on the context.

Example 3: Orthogonality of the Hadamard Basis

Show that the Hadamard basis states $|+\rangle$ and $|-\rangle$ are orthogonal.

Proof. Recall:

$$|+\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} \qquad |-\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$$

Orthogonal means that their inner product is zero. So, we need to compute $\langle +|-\rangle$ and show that it equals zero:

$$\langle +|-\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\end{bmatrix} \cdot \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix} = \frac{1}{2}(1-1) = 0$$

Therefore, $|+\rangle$ and $|-\rangle$ are orthogonal. QED

Example 4: Change of Basis from Standard $\leftrightarrow$ Hadamard

Given a qubit state in the standard basis:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

We want to express it in the Hadamard basis:

$$|\psi\rangle = x\,|+\rangle + y\,|-\rangle$$

Proof. Using the definitions of $|+\rangle$ and $|-\rangle$:

$$|+\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} \quad |-\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$$

We can write $|\psi\rangle$ in terms of $|+\rangle$ and $|-\rangle$:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle = a \cdot \begin{bmatrix}1\\0\end{bmatrix} + b \cdot \begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}a\\b\end{bmatrix}$$

Now, we express $|\psi\rangle$ as a linear combination of $|+\rangle$ and $|-\rangle$:

$$|\psi\rangle = x\,|+\rangle + y\,|-\rangle$$

Substituting the definitions of $|+\rangle$ and $|-\rangle$:

$$= x \cdot \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} + y \cdot \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$$

Combining the terms:

$$= \frac{1}{\sqrt{2}}\begin{bmatrix}x+y\\x-y\end{bmatrix}$$

We want this to equal $\begin{bmatrix}a\\b\end{bmatrix}$, so we set:

$$\frac{1}{\sqrt{2}}\begin{bmatrix}x+y\\x-y\end{bmatrix} = \begin{bmatrix}a\\b\end{bmatrix}$$

This gives us the system of equations:

$$x + y = \sqrt{2}a, \qquad x - y = \sqrt{2}b$$

Adding the two equations:

$$2x = \sqrt{2}(a + b) \implies x = \frac{a + b}{\sqrt{2}}$$

Subtracting the second from the first:

$$2y = \sqrt{2}(a - b) \implies y = \frac{a - b}{\sqrt{2}}$$

It follows that:

$$|\psi\rangle = x\,|+\rangle + y\,|-\rangle = \frac{a + b}{\sqrt{2}}\,|+\rangle + \frac{a - b}{\sqrt{2}}\,|-\rangle$$

QED

Example 5: Orthogonal State Construction

Given:

$$|v\rangle = a\,|0\rangle + b\,|1\rangle$$

Define:

$$|w\rangle = \bar{b}\,|0\rangle - \bar{a}\,|1\rangle$$

Show that $|v\rangle$ and $|w\rangle$ are orthogonal.

Proof. To show that $|v\rangle$ and $|w\rangle$ are orthogonal, we need to compute their inner product $\langle w|v\rangle$ and show that it equals zero. First, we write the inner product converting $|w\rangle$ to its corresponding bra:

$$|w\rangle = \bar{b}\,|0\rangle - \bar{a}\,|1\rangle \implies \langle w| = b\,\langle 0| - a\,\langle 1|$$

Now we compute:

$$\langle w|v\rangle = (b\,\langle 0| - a\,\langle 1|) \cdot (a\,|0\rangle + b\,|1\rangle)$$

Expanding this expression:

$$= ba\,\langle 0|0\rangle + bb\,\langle 0|1\rangle - aa\,\langle 1|0\rangle - ab\,\langle 1|1\rangle$$

Using the orthonormality of the basis states:

$$\langle 0|0\rangle = \langle 1|1\rangle = 1, \quad \langle 0|1\rangle = \langle 1|0\rangle = 0$$

We get:

$$\langle w|v\rangle = ba\cancelto{1}{\langle 0|0\rangle} + \cancel{bb\,\langle 0|1\rangle} - \cancel{aa\,\langle 1|0\rangle} - ab\cancelto{1}{\langle 1|1\rangle} = ba - ab$$

Since a and b are complex numbers, multiplication is commutative, so $ba = ab$. Therefore:

$$\langle w|v\rangle = ba - ab = 0$$

Hence, $|v\rangle$ and $|w\rangle$ are orthogonal. QED

2.3 Single Qubit Measurement

After introducing the mathematical representation of a qubit, we now describe how information is extracted from it through **measurement**.

What is *Single Qubit Measurement*?

We have a qubit:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

Where a and b are complex numbers such that $|a|^2 + |b|^2 = 1$. Measurement answers the question: "*which classical value do we observe?*".

A measurement takes a **quantum state** as input and produces a **classical outcome** as output. In the case of a single qubit, the possible outcomes are 0 and 1.

Measurement rule (computational basis)

The probability of an outcome is the **squared magnitude of the coefficient of the basis vector**:

$$\mathrm{P}(0) = |a|^2, \quad \mathrm{P}(1) = |b|^2 \quad \text{(in the computational basis)}$$

But which coefficients? This is where the **basis** comes in. A state can be written in different ways depending on the basis. For example, take:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

- If we measure in basis $\{|0\rangle, |1\rangle\}$:
  - Outcome 0 with probability $|a|^2$
  - Outcome 1 with probability $|b|^2$
- If we measure in basis $\{|+\rangle, |-\rangle\}$:
  - Outcome $+$ with probability $|x|^2$
  - Outcome $-$ with probability $|y|^2$

The **coefficients depend on the chosen basis**. Let's analyze how a basis change affects the coefficients.

What is a basis *really*? Any measurement device (e.g., a polarizer, page 12) has two preferred states (i.e., the states that it can distinguish). A basis is not just a mathematical construct; it corresponds to **what our measurements device can distinguish**. For example:

- Computational basis, the device distinguishes: *are you* $|0\rangle$ *or* $|1\rangle$*?*
- Hadamard basis, the device distinguishes: *are you* $|+\rangle$ *or* $|-\rangle$*?*

These are **different physical equations**.

The math behind the measurement. Measurement is projecting the state vector onto basis vectors. Forget qubits for a second and imagine a 2D real vector $\vec{v}$, and choose axes:

- **Standard axes**: we project onto x-axis and y-axis. We get components: v_x and v_y.
- **Rotated axes**: we rotate axes by $45°$ and project onto new axes the same vector $\vec{v}$. We get new components: v_+ and v_-.

Different numbers, same vector. The components are **not intrinsic properties of the vector**, but depend on the chosen axes. Back to qubits, **measurement projects the vector $|\psi\rangle$ onto the measurement axes** (i.e., the basis vectors). So $\langle u|\psi\rangle$ is the projection of $|\psi\rangle$ onto $|u\rangle$, and $|\langle u|\psi\rangle|^2$ is the probability of observing outcome u.

Example 6: Analogy - Measurement as projecting a vector onto different axes

Consider a classical 2D vector pointing diagonally:

$$\vec{v} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

Case 1: Standard axes (horizontal/vertical). If we describe the vector using the standard basis:

$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\},$$

Then the vector has equal components along both axes:

$$\vec{v} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- Projection on horizontal axis: $\frac{1}{\sqrt{2}}$
- Projection on vertical axis: $\frac{1}{\sqrt{2}}$

Case 2: Rotated axes (diagonal basis). Now rotate the coordinate system by $45°$ and use the basis:

$$\left\{ \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$$

In this basis, the same vector becomes:

$$\vec{v} = 1 \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 0 \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

- Projection on first axis: 1
- Projection on second axis: 0

Connection with qubits. A qubit behaves exactly in the same way:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = |+\rangle$$

- In basis $\{|0\rangle, |1\rangle\}$: equal projections $\Rightarrow$ probabilistic measurement.
- In basis $\{|+\rangle, |-\rangle\}$: full projection on $|+\rangle$ $\Rightarrow$ deterministic measurement.

The state does not change. What changes is the set of axes (basis) onto which the state is projected. Since measurement probabilities are given by these projections, the outcome depends on the chosen basis.

Given the state:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

- If we measure in the **computational basis**, we get:
  - Outcome 0 with probability $\left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$
  - Outcome 1 with probability $\left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$

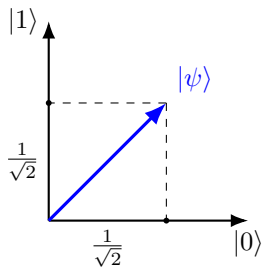


Figure 4: Measurement as projection onto basis vectors.

- If we measure in the **Hadamard basis**, we get:
  - Outcome $+$ with probability $|\langle +|\psi\rangle|^2 = 1$
  - Outcome $-$ with probability $|\langle -|\psi\rangle|^2 = 0$

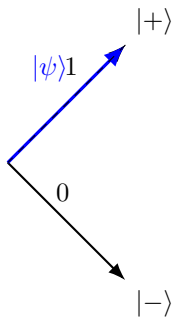


Figure 5: Measurement in the Hadamard basis. The state $|\psi\rangle$ is aligned with $|+\rangle$, so we get outcome $+$ with probability 1. The projection onto $|-\rangle$ is zero, so we get outcome $-$ with probability 0. The axes are rotated by 45° compared to the computational basis.

What happens to the state after measurement? The state **collapses** to the observed outcome. For example, if we measure $|\psi\rangle = a\,|0\rangle + b\,|1\rangle$ in the computational basis and observe 0, the state collapses to $|0\rangle$. If we observe 1, the state collapses to $|1\rangle$.

Two possible situations

As we have seen in page 20, the same state can yield different measurement outcomes depending on the chosen basis. This leads to two important situations:

- **Probabilistic measurement**: If the **state is not aligned with the measurement basis**, we get a probabilistic outcome. For example:

  $$|\psi\rangle = \frac{1}{\sqrt{2}}\,|0\rangle + \frac{1}{\sqrt{2}}\,|1\rangle$$

  Measured in the computational basis gives 0 or 1 with equal probability.

- **Deterministic measurement**: If the **state is aligned with one of the basis vectors**, we get a deterministic outcome. For example:

  $$|\psi\rangle = |+\rangle$$

  Measured in the Hadamard basis gives $+$ with probability 1 and $-$ with probability 0.

When should we prefer one situation to another? It depends on the task. For example, if we want to encode a classical bit, we prefer deterministic measurement (e.g., $|0\rangle$ or $|1\rangle$). If we want to create superposition states for quantum algorithms, we might want probabilistic measurement (e.g., $|+\rangle$).

Now, if the same quantum state can lead to either deterministic or probabilistic measurement outcomes depending on the chosen basis, we can ask: "*how can be explained this phenomenon?*". It means that a qubit can simultaneously have non-zero components along multiple basis states. This is a fundamental property of quantum mechanics, and it is what allows quantum computers to perform certain computations more efficiently than classical computers. It is called **superposition**, and we will discuss it in the next section.

2.4 Superposition

We have already seen that:

1. **A qubit is represented as a vector**:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

   Where a and b are complex numbers, and $|0\rangle$ and $|1\rangle$ are the basis states of the qubit.

2. **Measurement probabilities depend on the chosen basis**:
   - In the computational basis $\{|0\rangle, |1\rangle\}$, the probabilities of measuring $|0\rangle$ and $|1\rangle$ are $|a|^2$ and $|b|^2$, respectively.
   - In another basis, such as the Hadamard basis $\{|+\rangle, |-\rangle\}$, the probabilities of measuring $|+\rangle$ and $|-\rangle$ depend on different coefficients, which can be calculated by expressing $|\psi\rangle$ in terms of the new basis states.

3. **The same state can lead to deterministic or probabilistic outcomes depending on the measurement basis**. For example, if $|\psi\rangle = |0\rangle$, measuring in the computational basis always yield $|0\rangle$ (deterministic):

$$|0\rangle = 1\,|0\rangle + 0\,|1\rangle = |0\rangle$$

   Whereas measuring in the Hadamard basis yields $|+\rangle$ or $|-\rangle$ with equal probability (probabilistic):

$$|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$$

These observations raise an essential question: "*why can a single state exhibit such different behaviors under different measurements?*". The answer is **superposition**.

What is Superposition?

Superposition is a fundamental principle of quantum mechanics that **applies to all quantum systems**, not just qubits.

Definition 1: Superposition

Superposition means that a **system can exist in several possible states at the same time until a measurement is made**. It is **basis-dependent**, meaning a state may be a superposition in one basis but not in another, for example:

$$|v\rangle = \frac{1}{\sqrt{2}}\,|0\rangle + \frac{1}{\sqrt{2}}\,|1\rangle$$

Is a superposition with respect to the computational basis $\{|0\rangle, |1\rangle\}$, but it is **not** a superposition in the Hadamard basis $\{|+\rangle, |-\rangle\}$, where it can

be written as:

$$|v\rangle = 1\,|+\rangle + 0\,|-\rangle = |+\rangle$$

Formally, let $\{|u\rangle, |u^{\perp}\rangle\}$ be an orthonormal basis of a single-qubit Hilbert space. A qubit is said to be in **superposition** with respect to this basis if it can be written as:

$$|\psi\rangle = a\,|u\rangle + b\,|u^{\perp}\rangle$$

Where both coefficients a and b are non-zero and satisfy $|a|^2 + |b|^2 = 1$ (normalization condition). If one of the coefficients is zero, the state coincides with one of the basis vectors and the measurements in that basis becomes deterministic, thus it is not a superposition.

More in general, a quantum state $|\psi\rangle$ in a system with multiple possible states can exist in a **linear combination** of these states:

$$|\psi\rangle = c_1\,|\psi_1\rangle + c_2\,|\psi_2\rangle + \cdots + c_n\,|\psi_n\rangle$$

Where:

- $|\psi_1\rangle, |\psi_2\rangle, \ldots, |\psi_n\rangle$ are **basis states** of the system.
- $c_1, c_2, \ldots, c_n$ are **complex probability amplitudes**.
- The system is in **all states at the same time**, with the probability of measuring each state given by $|c_i|^2$.
- The state is **normalized**, meaning:

$$|c_1|^2 + |c_2|^2 + \cdots + |c_n|^2 = 1$$

Example 7: Single-Particle Systems (Quantum Mechanics)

In general quantum mechanics, a particle can exist in **multiple positions simultaneously** as a wave function $\Psi(x)$:

$$|\Psi\rangle = \int \Psi(x)\ |x\rangle\ \mathrm{d}x$$

- The particle is in a **superposition of all possible positions** $|x\rangle$.
- Measurement collapses the wave function to a single position.

How Can a Particle Exist in Multiple States at Once? This question touches the heart of quantum mechanics, where our everyday intuition breaks down. The **key idea** is that a **quantum particle** is not just a tiny ball, it **is a wave function that spreads across multiple possibilities at once**.

1. **Quantum Particles Are Waves, Not Just Points**. In classical physics, we think of particles as tiny, solid objects, like a small ball that always has a precise position and velocity.

   In quantum mechanics, however, particles behave more like waves. These waves are described by a wave function $\Psi(x)$, which represents the **probability of finding the particle at different locations**.

   The key idea is:

   - The **wave function spreads across space**, meaning the **particle does not have a single location** before measurement.
   - Instead, it exists in a superposition of all possible locations.

2. **The Double-Slit Experiment: Proof That a Particle Can Be in Two Places at Once**. The double-split experiment was first performed by Thomas Young in 1801 to demonstrate the **wave nature of light**. Later, in the 20th century, it was repeated with **electrons**, **atoms**, and even **large molecules**, revealing that **matter itself exhibits wave-like behavior**. The classical expectation for the double-slit experiment is:

   - **Classical Particles** (e.g., tiny balls). Imagine firing small balls at a barrier with two slits:
     - Each ball passes through **either** slit 1 **or** slit 2.
     - On a screen behind the slits, we would observe **two bright bands**, corresponding to the two slits.

     The result is a simple pattern of two lines, no interference.
   - **Classical Waves** (e.g., Water or Light). If instead we send **waves** (like water waves or classical light):
     - Each wave passes through **both slits simultaneously**.
     - The waves emerging from the slits **interfere**: **constructive interference** creates bright bands, while **destructive interference** creates dark bands.

     The result is an **interference pattern** of alternating bright and dark bands.

   The quantum reality is even more surprising:

   - **Quantum Particles** (e.g., photons or electrons). When the experiment is performed with **single quantum particles** (photons or electrons), something remarkable happens:
     (a) **Particles are emitted one at a time**.
     (b) Each particle is detected as a **localized point** on the screen.
     (c) After many particles are detected, an **interference pattern gradually emerges**.

     This means that each particle behaves as if it passes through **both slits simultaneously**. The particle is described by a **wave function**, representing a **superposition of paths**:

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(|\text{slit } 1\rangle + |\text{slit } 2\rangle\right)$$

The interference pattern arises from the **coherent combination** of these two possibilities.

Suppose we place detectors at the slits to determine **which slit** the particle passes through. The interference pattern **disappears** and the screen now shows **two bands**, just like classical particles. This is because measuring the particle's path **entangles** (i.e., correlates) it with the environment (the detector), leading to **decoherence** (loss of quantum coherence). The superposition:

$$\frac{1}{\sqrt{2}}\left(|\text{slit } 1\rangle + |\text{slit } 2\rangle\right)$$

Becomes effectively a **classical mixture**:

$$\frac{1}{2}|\text{slit } 1\rangle\langle\text{slit } 1| + \frac{1}{2}|\text{slit } 2\rangle\langle\text{slit } 2|$$

Without coherence, **interference cannot occur**.

In summary, the double-slit experiment demonstrates the principle of quantum superposition. When quantum particles such as photons or electrons are sent one at a time through two slits, they form an interference pattern on a detection screen. This indicates that each particle behaves as if it travels through both slits simultaneously, corresponding to the superposition:

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(|\text{slit } 1\rangle + |\text{slit } 2\rangle\right)$$

If a measurement is performed to determine which slit the particle passes through, the interference pattern disappears due to decoherence, and the system behaves like a classical probabilistic mixture.

Double Slit Experiment explained! by Jim Al-Khalili

3. **Quantum Superposition: More Than Just Probability**. A common misconception is that a particle in superposition is just **an unknown state**, like a coin that is either heads or tails, but we just don't know which. This is wrong, because quantum superposition is much deeper.

   A quantum state is a **combination of all possibilities**. *Until measurement*, the system is in **all possible states at once**.

   Mathematically, for an electron in **two locations** x_1 and x_2:

$$|\psi\rangle = a|x_1\rangle + b|x_2\rangle$$

   - The electron is **literally in both places simultaneously**.
   - The coefficients a and b are **complex numbers representing the probability amplitudes**.

- **Interference** between these amplitudes **creates quantum effects that cannot be explained by classical probability**.

4. **Superposition in Quantum Computing**. Quantum computing **directly uses** the fact that a particle can be in multiple states at once.

   - A **classical bit** can only be **0 or 1**.
   - A **qubit (quantum bit)** can be in a superposition:

   $$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

   This means a quantum computer can **perform many calculations simultaneously**. Therefore, **superposition allows quantum computers to process information exponentially faster than classical computers for certain tasks**.

5. **Why Don't We See Superposition in Everyday Life?** In our daily experience, objects are **not in multiple states at once** because of a process called **quantum decoherence**.

   Quantum superposition is **fragile**. When a quantum system interacts with the environment (air, light, etc.), the superposition **collapses into one definite state**. This is why large objects (like humans or cars) do not appear in multiple places at once.

   However, experiments (like the Double-Slit experiment) confirm that superposition is real at microscopic scales (electrons, photons, atoms, and even molecules).

The key **properties of Superposition** are:

1. **Linearity**: Any combination of valid quantum states is also a valid quantum state.

2. **Interference**: Quantum states in superposition can interfere, leading to constructive or destructive interference.

3. **Measurement Collapse**: When measured, the superposition collapses into a single outcome.

4. **Phase Information**: Unlike classical probabilities, quantum superpositions include complex phases that affect interference patterns.

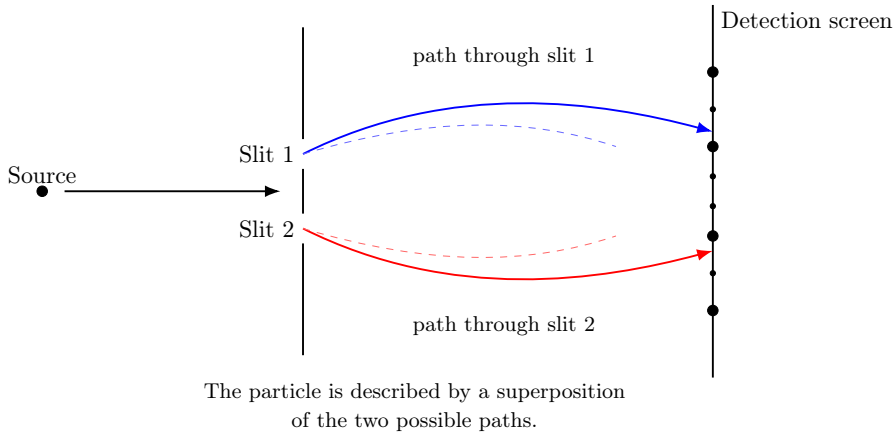


Figure 6: Double-slit experiment, superposition of paths. This figure illustrates the **quantum state of the particle before detection**. Instead of choosing a single path, the particle is described by a **superposition**:

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(|\text{slit } 1\rangle + |\text{slit } 2\rangle\right)$$

The particle is **not** traveling through only one slit, instead, it is described by a **coherent combination** of both possibilities. The diagram emphasizes the **quantum state** rather than the final inference pattern.

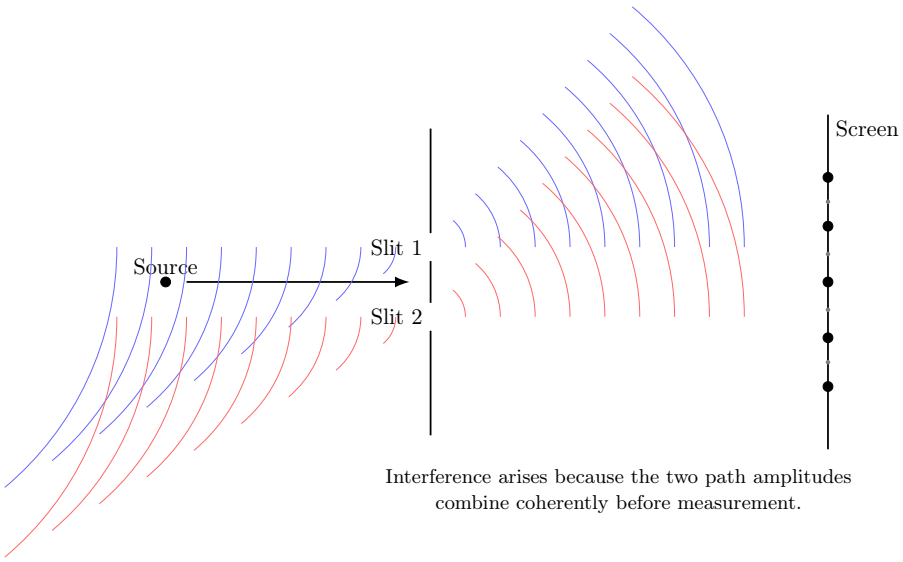


Figure 7: Wave-like representation of the double-slit experiment, interference of amplitudes. This figure focuses on the **wave nature** of the quantum particle. Each slit acts as a **coherent source**, producing wavefronts that overlap and interfere. If $A_1(x)$ and $A_2(x)$ are the probability amplitudes associated with the two slits, the probability of detecting the particle at position x on the screen is:

$$P(x) = |A_1(x) + A_2(x)|^2 = |A_1(x)|^2 + |A_2(x)|^2 + 2\mathrm{Re}\left(A_1^*(x)A_2(x)\right)$$

The last term is the **inference term**, which can be positive (bright fringes, **constructive interference**) or negative (dark fringes, **destructive interference**). This illustrates how the superposition of paths leads to the characteristic interference pattern observed in the experiment.

2.5 Information Carried by a Single Qubit

This section addresses a fundamental question: ***if a qubit can exist in infinitely many states, how much information can it actually convey?*** A general single-qubit state is given by:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle \qquad a, b \in \mathbb{C}, \quad |a|^2 + |b|^2 = 1$$

Because a and b are complex numbers, a qubit can be prepared in **infinitely many distinct states**. This might suggest that a qubit can encode an infinite amount of information. However, this is **not** the case. When we perform a **measurement**, we obtain only a **single classical outcome** (either 0 or 1). Therefore, a single qubit can yield at most one classical bit of information when measured. This apparent paradox, **infinite possible states but limited measurable information**, is one of the most important conceptual aspects of quantum information theory.

Why can't we determine the Amplitudes a and b of a single qubit state?

Suppose we are given a qubit in an unknown state $|\psi\rangle$. Can we determine a and b from a single measurement? The answer is **no**, because:

- ✗ **Measurement is probabilistic**. A single measurement provides only one outcome (0 or 1), which is insufficient to determine the amplitudes.
- ✗ **Measurement disturbs the state**. After measurement, the qubit collapses to the observed basis state, preventing further measurements on the original state.
- ✗ **State estimation requires many copies**. To estimate a and b, we need **quantum state tomography**, which relies on performing measurements on **many identically prepared qubits**.

This **irreversible collapse** means that the original information encoded in the amplitudes is generally lost after measurement.

The No-Cloning Theorem

One might think of copying the qubit before measurement to extract more information. However, quantum mechanics forbids this through the **No-Cloning Theorem**.

Theorem 1 (No-Cloning). *It is impossible to create an identical copy of an arbitrary unknown quantum state.*

Proof. Assume there exists a unitary operator U that can clone any state $|\psi\rangle$:

$$U\,|\psi\rangle\,|0\rangle = |\psi\rangle\,|\psi\rangle \quad \forall\,|\psi\rangle$$

For two arbitrary states $|\psi\rangle$ and $|\phi\rangle$, linearity of quantum mechanics implies:

$$U\left(\alpha\,|\psi\rangle + \beta\,|\phi\rangle\right)|0\rangle = \alpha\,|\psi\rangle\,|\psi\rangle + \beta\,|\phi\rangle\,|\phi\rangle$$

Which is **not equal** to:

$$(\alpha |\psi\rangle + \beta |\phi\rangle)(\alpha |\psi\rangle + \beta |\phi\rangle) = \alpha^2 |\psi\rangle |\psi\rangle + \alpha\beta |\psi\rangle |\phi\rangle + \alpha\beta |\phi\rangle |\psi\rangle + \beta^2 |\phi\rangle |\phi\rangle$$

This contradiction proves that perfect cloning is impossible, since the two expressions cannot be equal for all $|\psi\rangle$ and $|\phi\rangle$. QED

Since we cannot clone a qubit, we cannot create multiple copies to determine its exact state from a single instance.

Feature	Classical Bit	Qubit
Possible states	0 or 1	Qubit
Info from measurement	1 bit	At most 1 bit
Measurement disturbance	No	Yes
Ability to copy	Yes	No (No-Cloning)
State determination	Single observation	Requires many identical copies

Table 1: Comparison of classical bits and qubits in terms of information content and measurement properties.

2.6 State Space of a Single-Qubit System

The **State Space** of a physical system is the set of all possible states that the system can occupy. For a **single qubit**, the state space consists of all vectors of the form:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

- $a, b \in \mathbb{C}$ are **complex probability amplitudes**.
- $|0\rangle$ and $|1\rangle$ form an **orthonormal basis** (the computational basis).
- The **normalization condition** must hold: $|a|^2 + |b|^2 = 1$. This ensures that the total probability of all measurement outcomes is equal to 1.

Degrees of Freedom of a Qubit

At first glance, the coefficients a and b appear to provide **4 real parameters**:

$$\begin{aligned} a = a_r + ia_i \quad &\Rightarrow \quad a_r, a_i \quad \text{(2 parameters)} \\ b = b_r + ib_i \quad &\Rightarrow \quad b_r, b_i \quad \text{(2 parameters)} \end{aligned}$$

So initially, it seems that a qubit has **four degrees of freedom**. However, not all of these correspond to physically distinguishable states.

- **Normalization constraint**. Quantum states must be normalized, which imposes the condition:

  $$|a|^2 + |b|^2 = a_r^2 + a_i^2 + b_r^2 + b_i^2 = 1$$

  Geometrically, it defines the surface of a **unit sphere in four dimension** (also known as a 3-sphere, S^3). This constraint effectively removes **one degree of freedom**, since the amplitudes must lie on the surface of a 3-sphere in 4-dimensional space.

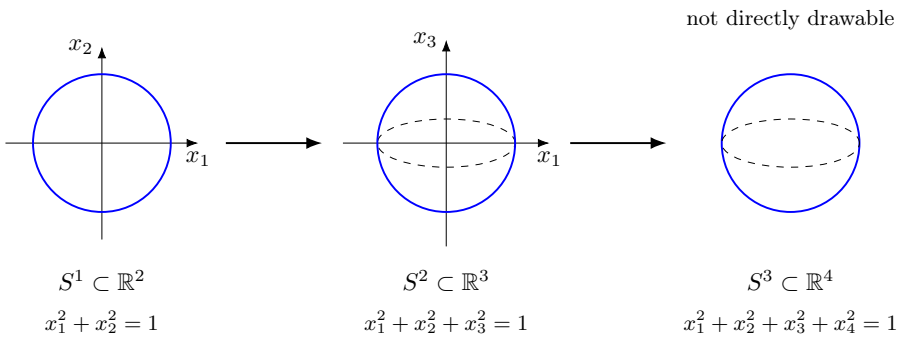


Figure 8: Analogy between the unit circle S^1, the ordinary sphere S^2, and the 3-sphere S^3. The normalized qubit state lives on S^3 because it is specified by four real coordinates constrained by one equation. In this figure, we can only visualize S^1 and S^2, while S^3 cannot be directly drawn in three-dimensional space, so we represent it with a dashed ellipse to indicate that it is a higher-dimensional object.

- **Global Phase Equivalence**. Two states that differ only by a **global phase** represent the **same physical qubit**:

$$|\psi'\rangle = e^{i\phi}|\psi\rangle$$

For example:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad e^{i\pi/3}|\psi\rangle = \frac{e^{i\pi/3}}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Why is it physically irrelevant? Measurement probabilities are given by:

$$P(i) = |\langle i|\psi\rangle|^2$$

If we multiply the state by a global phase:

$$\left|\langle i|e^{i\gamma}\psi\rangle\right|^2 = \left|e^{i\gamma}\langle i|\psi\rangle\right|^2 = |\langle i|\psi\rangle|^2$$

Because $\left|e^{i\gamma}\right| = 1$ for any real γ.

Mathematical proof that $\left|e^{i\gamma}\right| = 1$. We can express $e^{i\gamma}$ using Euler's formula:

$$e^{i\gamma} = \cos(\gamma) + i\sin(\gamma)$$

The magnitude of a complex number $z = x + iy$ is given by:

$$|z| = \sqrt{x^2 + y^2}$$

Applying this to $e^{i\gamma}$:

$$\begin{aligned}
\left|e^{i\gamma}\right| &= \sqrt{(\cos(\gamma))^2 + (\sin(\gamma))^2} \\
&\downarrow \text{using the trigonometric identity} \\
&= \sqrt{\cos^2\gamma + \sin^2\gamma} \\
&= \sqrt{1} \\
&= 1
\end{aligned}$$

QED

Therefore, **no experiment can distinguish between** $|\psi\rangle$ **and** $e^{i\phi}|\psi\rangle$, since **they yield the same measurement probabilities** (due to $\left|e^{i\phi}\right| = 1$).

How does this remove a degree of freedom? Unlike normalization, the global phase does **not** impose a constrain equation. Instead, it introduces a **redundancy** in our description:

- Many different vectors in S^3 correspond to the **same physical state**.
- All vectors of the norm $e^{i\gamma}|\psi\rangle$ are considered **physically equivalent** to $|\psi\rangle$.

Mathematically, we are **identifying** all these vectors as a single point. This process is called forming a **quotient space**: the set of physically distinct pure qubit states is $S^3 \setminus S^1 \cong S^2$, which is a 2-dimensional surface (the Bloch sphere, see Figure 8, page 32).

After accounting for these constraints, we find that a single qubit state is fully described by **two real parameters** (two degrees of freedom). Therefore, any qubit state can be written in the form:

1. A normalized qubit can be written as:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle \qquad a, b \in \mathbb{C} \quad |a|^2 + |b|^2 = 1$$

2. Any complex number can be written using its **magnitude** and **phase**:

$$a = |a|\,e^{i\alpha} \qquad b = |b|\,e^{i\beta}$$

Where $|a|$ and $|b|$ are non-negative real numbers, and α and β are real numbers representing the phases. Substituting these into the expression for $|\psi\rangle$ gives:

$$|\psi\rangle = \underbrace{|a|\,e^{i\alpha}}_{a}\,|0\rangle + \underbrace{|b|\,e^{i\beta}}_{b}\,|1\rangle$$

3. We can factor out the common global phase $e^{i\alpha}$:

$$|\psi\rangle = e^{i\alpha}\left(|a|\,|0\rangle + |b|\,e^{i(\beta-\alpha)}\,|1\rangle\right)$$

Since the global phase $e^{i\alpha}$ has **no physical effect**, we can omit it, leading to:

$$|\psi\rangle = |a|\,|0\rangle + |b|\,e^{i\phi}\,|1\rangle$$

Where we define also the **Relative Phase** $\phi = \beta - \alpha$. At this point, the state is characterized by two real magnitudes $|a|$ and $|b|$, and one relative phase ϕ. Mathematically, if:

$$a = |a|\,e^{i\alpha} \qquad b = |b|\,e^{i\beta}$$

Then:

$$\text{Relative Phase} = \phi = \arg(b) - \arg(a) = \beta - \alpha \tag{1}$$

4. From normalization:

$$|a|^2 + |b|^2 = 1$$

We recognize a familiar trigonometric identity:

$$\cos^2(x) + \sin^2(x) = 1$$

Therefore, we can parametrize the magnitudes using a single angle θ:

$$|a| = \cos\left(\frac{\theta}{2}\right) \qquad |b| = \sin\left(\frac{\theta}{2}\right)$$

With $0 \leq \theta \leq \pi$ to ensure that $|a|$ and $|b|$ are non-negative. The reason for using $\frac{\theta}{2}$ instead of θ will become clear when we discuss the Bloch sphere representation (Figure 9, page 36).

5. Substituting the parametrized magnitudes into the state:

$$|\psi\rangle = \underbrace{\cos\left(\frac{\theta}{2}\right)}_{|a|}|0\rangle + e^{i\phi}\underbrace{\sin\left(\frac{\theta}{2}\right)}_{|b|}|1\rangle \tag{2}$$

This is the **canonical parametrization** of a single qubit state.

Regardless of the specific values of θ and ϕ in Equation 2:

- θ: controls the relative weights of $|0\rangle$ and $|1\rangle$ in the superposition, determining the probabilities of measuring each state.
- ϕ: controls the relative phase between the $|0\rangle$ and $|1\rangle$ components. While it does not affect measurement probabilities, it is crucial for interference effects and the behavior of the qubit under quantum gates (which we will discuss later).

Bloch Sphere Representation of a Qubit

After normalization and removal of the physically irrelevant global phase, a pure single-qubit state depends on only two real parameters. Therefore, it is natural to represent it geometrically as a point on a two-dimensional surface. This surface is the **Bloch sphere**.

Any pure qubit can be written in the form:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle\,,$$

Where:

- $\theta \in [0, \pi]$ is the **polar angle**,
- $\phi \in [0, 2\pi)$ is the **Azimuthal Angle**, representing the **relative phase**.

This expression gives a one-to-one correspondence between pure qubit states and points on the surface of a unit sphere:

- $|0\rangle$ corresponds to the north pole, since $\theta = 0$:

$$|0\rangle = \cos\left(\frac{0}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{0}{2}\right)|1\rangle = 1\cdot|0\rangle + 0\cdot|1\rangle = |0\rangle$$

- $|1\rangle$ corresponds to the south pole, since $\theta = \pi$:

$$|1\rangle = \cos\left(\frac{\pi}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\pi}{2}\right)|1\rangle = 0\cdot|0\rangle + e^{i\phi}\cdot 1\cdot|1\rangle = e^{i\phi}|1\rangle \Rightarrow |1\rangle$$

Attention! $e^{i\phi}|1\rangle$ becomes $|1\rangle$ after **removing the global phase** (physically unobservable), so $|1\rangle$ is indeed at the south pole. In general, all states of the form $e^{i\phi}|1\rangle$ represent the **same physiical state** as $|1\rangle$. Hence, $|1\rangle$ corresponds to the **south pole** of the Bloch sphere.

- States on the equator correspond to equal-weight superpositions of $|0\rangle$ and $|1\rangle$.

In particular:

$$\text{Hadamard} \quad |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle),$$

$$\text{Imaginary} \quad |i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle), \quad |-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle),$$

All lie on the equator and differ only by their relative phase.

The Bloch sphere provides an intuitive visualization of a qubit: moving along the vertical direction changes the relative weights of $|0\rangle$ and $|1\rangle$, while rotating around the vertical axis changes the relative phase between them.

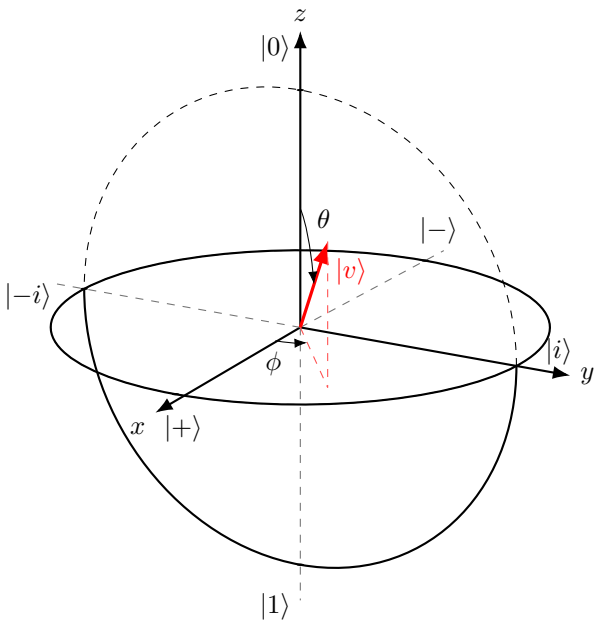


Figure 9: Bloch sphere representation of a pure qubit state. The red vector $|v\rangle$ represents a generic qubit state. Measuring $|v\rangle$ along the z-axis gives probabilities determined by θ, while along x or y-axes reveals information about the relative phase ϕ.

On the Bloch sphere, the axes correspond to **three fundamental measurement bases** for a qubit. Each pair of opposite points represents the **eigenstates of one of the Pauli operators** (we will discuss the Pauli operators in more detail in 3.3.2.2, page 47):

- **z-axis**, **computational basis** $\{|0\rangle, |1\rangle\}$, eigenstates of σ_z. The states are:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \text{and} \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

  These states correspond to the classical bit values 0 and 1.

- $|0\rangle$: North pole of the Bloch sphere $(0, 0, 1)$.
- $|1\rangle$: South pole of the Bloch sphere $(0, 0, -1)$.

Measuring along the z-axis answer the question: "*is the qubit* $|0\rangle$ *or* $|1\rangle$*?*".

- **x-axis**, **Hadamard basis** $\{|+\rangle, |-\rangle\}$, eigenstates of σ_x. The states are:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \qquad \text{and} \qquad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

  - $|+\rangle$: $+x$ direction on the equator $(1, 0, 0)$, with $\phi = 0$ (no relative phase).
  - $|-\rangle$: $-x$ direction on the equator $(-1, 0, 0)$, with $\phi = \pi$ (relative phase of π between $|0\rangle$ and $|1\rangle$).

  These states represent **equal superpositions** of $|0\rangle$ and $|1\rangle$ with **relative phase** 0 and π.

- **y-axis**, **imaginary basis** $\{|i\rangle, |-i\rangle\}$, eigenstates of σ_y. The states are:

$$|i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle) \qquad \text{and} \qquad |-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle)$$

  - $|i\rangle$: $+y$ direction on the equator $(0, 1, 0)$, with $\phi = \frac{\pi}{2}$ (relative phase of $\frac{\pi}{2}$ between $|0\rangle$ and $|1\rangle$).
  - $|-i\rangle$: $-y$ direction on the equator $(0, -1, 0)$, with $\phi = -\frac{\pi}{2}$ (relative phase of $-\frac{\pi}{2}$ between $|0\rangle$ and $|1\rangle$).

  These states also have equal probabilities of measuring $|0\rangle$ or $|1\rangle$ in the computational basis, but they differ by a **relative phase of** $\pm\frac{\pi}{2}$. This phase is crucial for **interference effects** in quantum algorithms.

3 Single Qubit Gates

3.1 Operations on Qubits

Quantum computing relies on three essential operations:

1. **Logic Gates**: transform quantum states.
2. **Measurements**: extract classical information from qubits.
3. **Initialization Procedures**: prepare qubits in known starting states.

Logic Gates: Reversible Transformations

Quantum **Logic Gates** are the building blocks of quantum circuits. They **manipulate the state of qubits in a controlled and deterministic way**, analogous to classical logic gates (like AND or NOT), but with key quantum differences:

- **Operate on Single or Multiple Qubits**. They can affect one qubit (single-qubit gates) or multiple qubits simultaneously (multi-qubit gates):
  - **Single-Qubit Gates**: These gates, such as the Pauli-X, Y, Z, Hadamard (H), and Phase (S) gates, operate on a single qubit to change its state. We will explore these gates in detail in the next sections.
  - **Multi-Qubit Gates**: These gates, such as the CNOT (Controlled-NOT) and Toffoli gates, operate on multiple qubits and can create *entanglement* (we will discuss this in detail later) between them, which is a key resource in quantum computing.
- **Reversible Transformations**. Unlike many classical gates (e.g., AND), **all quantum gates are reversible**. Mathematically, this means they are represented by **unitary matrices** U, satisfying:

  $$U^\dagger U = UU^\dagger = I$$

  Where $U^\dagger$ is the conjugate transpose of U, and I is the identity matrix. Because of this reversibility, quantum gates do not lose information and the original state can be recovered by applying $U^\dagger = U^{-1}$ to the output state. This is a fundamental difference from classical logic gates, which can be irreversible (e.g., AND gate loses information about the inputs).

  Finally, this property is related to fundamental principles such as the **no-cloning theorem** (page 30) and the **no-deleting theorem** (we cannot delete quantum information arbitrarily, since it is preserved under unitary transformations).
- **State Evolution**. If a qubit is in the state:

  $$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

Then after applying a gate U, the new state becomes:

$$|\psi'\rangle = U|\psi\rangle = aU|0\rangle + bU|1\rangle$$

This **transformation is deterministic** and **preserves the normalization of the state** (i.e., $|a|^2 + |b|^2 = 1$ remains true after applying U). The specific effect of the gate depends on its matrix representation, which we will explore in the next sections.

We can think of quantum logic gates as **rotations on the Bloch sphere**, smoothly transforming the qubit's state without destroying its quantum nature.

Measurement: Irreversible Collapse of State

Measurement is the process of extracting information from a qubit. Unlike logic gates, measurement is **irreversible** and fundamentally probabilistic.

- **Irreversible Process**. Once a qubit is measured, its original quantum state is **lost**. This distinguishes measurement from unitary gate operations.
- **State Collapse**. Measuring a qubit in the computational basis $\{|0\rangle, |1\rangle\}$ causes the state:

  $$|\psi\rangle = a|0\rangle + b|1\rangle$$

  To **collapse** to:

  - $|0\rangle$ with probability $|a|^2$.
  - $|1\rangle$ with probability $|b|^2$.
- **Extraction of Classical Information**. The measurement outcome is a **classical bit** (0 or 1). After measurement, subsequent measurements in the same basis yield the same result with probability 1, since the state has collapsed to a definite state.

Measurement can be described using **projection operators**:

$$M_0 = |0\rangle\langle 0| \quad M_1 = |1\rangle\langle 1|$$

The probability of obtaining outcome k (where $k \in \{0, 1\}$) when measuring $|\psi\rangle$ is given by:

$$\begin{aligned}
p_k &= \langle\psi| M_k |\psi\rangle \\
&\downarrow \text{substitute } M_k = |k\rangle\langle k| \\
&= \langle\psi| (|k\rangle\langle k|) |\psi\rangle \\
&\downarrow \text{using the associativity of inner products} \\
&= (\langle\psi|k\rangle)(\langle k|\psi\rangle) \\
&\downarrow \text{since } \langle\psi|k\rangle = (\langle k|\psi\rangle)^* \\
&\quad \text{where * denotes complex conjugation} \\
&= (\langle k|\psi\rangle)^* (\langle k|\psi\rangle) \\
&\downarrow \text{which is the definition of squared magnitude, page 7} \\
&= |\langle k|\psi\rangle|^2
\end{aligned}$$

In other words, the probability of measuring outcome k is the **squared magnitude of the amplitude** of $|\psi\rangle$ along the direction $|k\rangle$. And the post-measurement state of the qubit, conditioned on obtaining outcome k, becomes:

$$|\psi_k\rangle = \frac{M_k |\psi\rangle}{\sqrt{p_k}}$$

Measurement is like **asking a question** to the qubit: "are you in state $|0\rangle$ or $|1\rangle$?". The answer is classical (i.e., 0 or 1), but the act of asking **changes the system**.

Example 1: Calculating Measurement Probabilities

Consider a qubit in the state:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

With $|a|^2 + |b|^2 = 1$.

- Probability of measuring 0. Using the projector:

  $$M_0 = |0\rangle\langle 0|$$

  We compute:

  $$\begin{aligned} p_0 &= \langle\psi| M_0 |\psi\rangle \\ &= \langle\psi|0\rangle\langle 0|\psi\rangle \\ &= (\langle\psi|0\rangle)(\langle 0|\psi\rangle) \\ &\downarrow \text{since } \langle\psi|0\rangle = (\langle 0|\psi\rangle)^* \\ &= |\langle 0|\psi\rangle|^2 \\ &= |a|^2 \end{aligned}$$

- Probability of measuring 1. Using the projector:

  $$M_1 = |1\rangle\langle 1|$$

  We compute:

  $$\begin{aligned} p_1 &= \langle\psi| M_1 |\psi\rangle \\ &= \langle\psi|1\rangle\langle 1|\psi\rangle \\ &= (\langle\psi|1\rangle)(\langle 1|\psi\rangle) \\ &\downarrow \text{since } \langle\psi|1\rangle = (\langle 1|\psi\rangle)^* \\ &= |\langle 1|\psi\rangle|^2 \\ &= |b|^2 \end{aligned}$$

Initialization: Preparing Known States

The **Initialization** is the process of preparing a qubit in a **known and controlled state**, typically $|0\rangle$ or $|1\rangle$. This step is essential because quantum algorithms require a well-defined starting point. It is implemented through various methods, including:

- **Physical Preparation**. Many quantum technologies naturally relax to the ground state $|0\rangle$. Additional gates (e.g., Pauli-X) can be applied to obtain $|1\rangle$ if needed.
- **Initialization via Measurement**. Measuring a qubit projects it into a basis state. If the outcome is not the desired one, a corrective gate can be applied. For example, if measurement yields $|1\rangle$ but $|0\rangle$ is needed, apply an X gate to flip it to $|0\rangle$.

Initialization is the **first step** in any quantum computation: (1) **prepare** qubits in known states, (2) **apply** them using quantum gates, and (3) **measure** to obtain classical results. It is analogous to **resetting a classical register** before starting a computation in classical computing.

3.2 Quantum Logic Gates Overview

In quantum computing, quantum logic gates are the primary tools used to **manipulate qubits**. These gates perform **unitary transformations** on the state of one or more qubits, meaning they **preserve the norm** of the quantum state and are **reversible**. The design of quantum algorithms and the execution of quantum circuits rely entirely on the sequential application of these gates.

Types of Quantum Gates

Quantum Gates can be classified into:

- **Single-Qubit Gates**: These gates operate on **individual qubits**, modifying their state in isolation.
- **Multiple-Qubit Gates**: These gates act on **two or more qubits simultaneously**, allowing for entanglement and complex correlations between qubits.

In this section we will focus on single qubit gates.

Mathematical Framework: Qubit as Vectors, Gates as Matrices

To describe quantum superposition mathematically, we require a framework that supports linear combinations of states, probabilities derived from inner products, complex amplitudes and phases, and linear state transformations. The mathematical structure satisfying all these requirements is a **complex Hilbert space**. Thus, a qubit is naturally represented as a **unit vector** in a two-dimensional complex Hilbert space $\mathbb{C}^2$.

Using the Dirac notation, the basis states $|0\rangle$ and $|1\rangle$ form an orthonormal basis for this space, satisfying $\langle 0|0\rangle = \langle 1|1\rangle = 1$ and $\langle 0|1\rangle = \langle 1|0\rangle = 0$. Any qubit state can be expressed as a linear combination of these basis states:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle \qquad \text{with } |a|^2 + |b|^2 = 1$$

Two state vectors that differ only by a global phase factor $e^{i\phi}$ represent the same physical qubit, since measurement probabilities remain unchanged (see page 33).

Choosing the computational basis $\{|0\rangle, |1\rangle\}$, we can represent these states as column vectors:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Therefore, the qubit state can be written as:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$$

This vector representation allows us to apply linear algebra techniques to analyze and manipulate qubits.

Once qubits are represented as vectors, transformations of these states must be **linear operators** acting on the vector space. In a chosen basis, linear operators are represented by **matrices**. To preserve normalization and probabilities, these matrices must be **unitary**:

$$U^\dagger U = UU^\dagger = I$$

Thus, quantum gates – the **building blocks of quantum circuits** – are represented as unitary 2×2 matrices that linearly transform the state vector:

$$|\psi'\rangle = U|\psi\rangle$$

Why Matrices? Because quantum mechanics is inherently linear, any physical evolution of a closed quantum system must be described by a **linear operator**. When expressed in a chosen basis (such as the computational basis $\{|0\rangle, |1\rangle\}$), this operator takes the form of a matrix.

Unitary: The Fundamental Constraint

A matrix U is **unitary** if it satisfies the condition:

$$U^\dagger U = UU^\dagger = I \tag{3}$$

Where:

- $U^\dagger$ is the **Hermitian transpose** (conjugate transpose) of U.
- I is the **identity matrix**.

This property ensures that $U^{-1} = U^\dagger$, meaning every quantum operation is **reversible**, and the transformation preserves the **norm** of the state vector. So, quantum gate **must be unitary** to ensure that the evolution of quantum states is consistent with the principles of quantum mechanics (i.e., linearity, reversibility, and preservation of probabilities).

A general single-qubit unitary matrix can be expressed as:

$$U = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \quad \text{with} \quad \alpha, \beta, \gamma, \delta \in \mathbb{C} \quad \text{and} \quad U^\dagger U = I \tag{4}$$

Why must gates be unitary?

1. **No-Cloning Theorem**: It is impossible to create an identical copy of an unknown qubit. Unitary transformations respect this by not duplicating information.
2. **No-Deleting Theorem**: We cannot delete quantum information arbitrarily. Since unitary gates are invertible, they do not erase information.
3. **Preservation of Probability**: The normalization condition $|a|^2 + |b|^2 = 1$ must hold after any operation, and only unitary matrices preserve this norm.

Physical Intuition

In classical circuits, logic gates like AND, OR, NOT process bits deterministically. In quantum circuits:

- Gates **rotate** the quantum state on the **bloch sphere**.
- These rotations correspond to **unitary transformations**.
- The **physical implementation** of quantum gates may vary across hardware (e.g., superconducting qubits, trapped ions), but **mathematically**, the **same unitary matrix describes the gate**.

Quantum gates do not correspond to physical gates in the classical sense. They are **abstract operations** realized by **controlled interactions** in quantum systems.

3.3 Main Single-Qubit Gates

3.3.1 Identity Gate (I)

The **Identity Gate**, denoted I, is the most elementary gate in quantum computing. It performs **no change** to the state of a qubit. Its action on a qubit is trivial, yet it **serves important roles in quantum circuits**, particularly when managing multi-qubit systems or timing synchronization.

Matrix Representation

The Identity gate is represented by the 2×2 **identity matrix**:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{5}$$

Applying the Identity gate to any qubit **leaves the state unchanged**. Formally, for a qubit in state $|\psi\rangle$:

$$I\,|\psi\rangle = |\psi\rangle$$

To better understand its effect, consider its action on the computational basis:

$$I\,|0\rangle = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$
$$I\,|1\rangle = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$$

Hence, the Identity gate **preserves the state** of the qubit, including any superposition or entangled state it may be in.

Physical Interpretation

Although it may appear useless at first glance, the Identity gate has **conceptual and practical utility**:

- It acts as a **placeholder** or **"*do nothing*" operation** when required by **circuit timing**.
- In **multi-qubit circuits**, it allows one **qubit to remain unchanged** while other qubits are operated on.
- In **matrix composition** of gates, it serves to **align dimensions** when performing **tensor products** (e.g., $I \otimes H$ applies Hadamard to the second qubit only).

Bloch Sphere Perspective

In the Bloch sphere (page 36), the **identity gate** represents a **rotation of zero degrees**, meaning the state vector remains exactly where it is. It is the neutral element of quantum transformations.

3.3.2 Pauli Gates

The **Pauli Gates** are a set of fundamental single-qubit quantum gates that play a central role in quantum computing. They are represented by the three **Pauli matrices** X, Y, and Z, and can be interpreted as the quantum analogues of basic operations on a qubit.

Each Pauli gate corresponds to a **rotation of π radians** around one of the principal axes of the Bloch sphere:

- X: rotation around the x-axis (bit flip).
- Z: rotation around the z-axis (phase flip).
- Y: rotation around the y-axis (bit and phase flip).

These operators are **unitary** ($U^\dagger U = UU^\dagger = I$), **Hermitian** ($U = U^\dagger$), and **self-inverse** ($U^2 = I$), and their eigenstates define the three fundamental measurement bases of a qubit. Together, they form the building blocks for describing single-qubit transformations and quantum dynamics. Also, they have eigenvalues ± 1 and satisfy the property:

$$X^2 = Y^2 = Z^2 = I$$

Meaning that applying the same Pauli gate twice returns the qubit to its original state.

3.3.2.1 Rotations Generated by Pauli Operators

In quantum mechanics, rotations of a qubit are generated by the Pauli operators. A rotation around an axis $\hat{n}$ by an angle θ is given by:

$$R_{\hat{n}}(\theta) = e^{-i\frac{\theta}{2}(\hat{n}\cdot\vec{\sigma})} \tag{6}$$

Where $\vec{\sigma} = (X, Y, Z)$ are the Pauli matrices. We will see those matrices in more detail in the next sections, but for now we can just note that they are the generators of rotations around the x, y, and z axes, respectively.

In particular:

$$R_x(\theta) = e^{-i\frac{\theta}{2}X} \tag{7}$$

$$R_y(\theta) = e^{-i\frac{\theta}{2}Y} \tag{8}$$

$$R_z(\theta) = e^{-i\frac{\theta}{2}Z} \tag{9}$$

Using the identity $\sigma^2 = I$, these can be expanded as:

$$R_\alpha(\theta) = \cos\left(\frac{\theta}{2}\right) I - i\sin\left(\frac{\theta}{2}\right)\sigma_\alpha \tag{10}$$

3.3.2.2 Pauli-X (NOT) Gate

The **Pauli-X Gate**, often called the **quantum NOT gate**, is one of the *fundamental* single-qubit quantum gates. It is denoted X and **corresponds to the σ_x Pauli matrix** in quantum mechanics. This gate **flips the state** of a qubit, **similar to how a classical NOT gate inverts a bit**.

Remark: Pauli Matrix

In quantum mechanics, the **Pauli matrices** are **a set of three** 2×2 **Hermitian and unitary matrices** that represent **spin operators** for spin-$\frac{1}{2}$ particles and are widely used in quantum computing to describe single-qubit gates. They are:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Key properties:

- Hermitian: $\sigma_i^\dagger = \sigma_i$ (observable-related)
- Unitary: $\sigma_i^\dagger \sigma_i = I$ (reversible transformations)
- Involution: $\sigma_i^2 = I$
- Non-commuting: $[\sigma_i, \sigma_j] = 2i\varepsilon_{ijk}\sigma_k$ (with ε the Levi-Civita symbol)

Matrix Representation

$$X = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{11}$$

This **matrix swaps** the $|0\rangle$ and $|1\rangle$ **basis states**:

$$X|0\rangle = |1\rangle \qquad \text{and} \qquad X|1\rangle = |0\rangle \tag{12}$$

Action on a General Qubit

Let's consider a qubit in the general state:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

Applying the X gate:

$$X|\psi\rangle = aX|0\rangle + bX|1\rangle = a|1\rangle + b|0\rangle = b|0\rangle + a|1\rangle$$

As a result, the **amplitudes** a and b are **swapped**.

Key Properties of X Gate

The Pauli-X gate satisfies several important mathematical properties:

1. **Unitary**. The X gate is unitary, meaning it preserves the inner product and is reversible:
$$XX^\dagger = I$$

2. **Hermitian**. The X gate is also Hermitian, which means it is equal to its own conjugate transpose:
$$X = X^\dagger$$

3. **Self-Inverse**. The X gate is its own inverse, which means that **applying it twice returns the qubit to its original state**:
$$X^{-1} = X \quad \Rightarrow \quad X^2 = I$$

Geometric Interpretation: Rotation on the Bloch Sphere

The Pauli-X gate corresponds to a **rotation around the x-axis by π radians** (180°) on the Bloch sphere, denoted as $X = R_x(\pi)$.

To understand this, recall the **general qubit state in spherical coordinates**:
$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

After a π rotation around the x-axis:

- $\theta \to \pi - \theta$
- $\phi \to -\phi$

This **rotation mirrors the qubit across the x-axis**, flipping its position between $|0\rangle$ and $|1\rangle$, and adjusting the phase accordingly.

Rotation Interpretation

From Equation 6 (page 46) and Equation 10 (page 46), a rotation around the x-axis by an angle θ is given by:
$$R_x(\theta) = \cos\left(\frac{\theta}{2}\right)I - i\sin\left(\frac{\theta}{2}\right)X$$

For $\theta = \pi$, we obtain:
$$R_x(\pi) = -iX$$

Therefore, the Pauli-X gate corresponds to a **rotation of π around the x-axis, up to a global phase**:
$$X \equiv R_x(\pi)$$

Since global phases are physically irrelevant, X and $R_x(\pi)$ represent the same physical transformation.

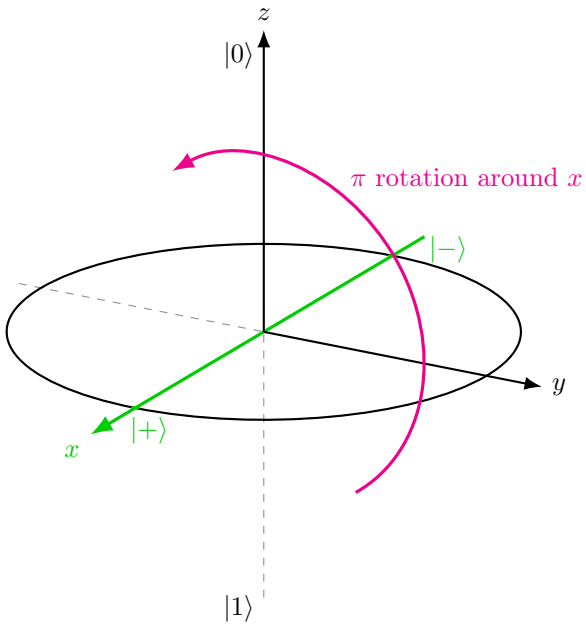


Figure 10: Eigenstates of the Pauli-X gate. Since $|+\rangle$ and $|-\rangle$ lie on the x-axis, they are not moved by a rotation around that axis. The state $|-\rangle$ only acquires a global phase -1.

Eigenstates

In linear algebra, an **eigenvector** of a matrix (or linear operator) A is a non-zero vector $|v\rangle$ that, when the operator is applied, changes only by a **scalar factor** called the **eigenvalue** λ:

$$A\,|v\rangle = \lambda\,|v\rangle$$

In quantum mechanics, we use the term **Eigenstate** instead of eigenvector. So, an eigenstate of an operator U satisfies:

$$U\,|v\rangle = \lambda\,|v\rangle$$

An **eigenstate** is a state that remains **unchanged in direction** after the operation is applied, except only its **magnitude** or **phase** may change by a factor of λ. For the Pauli-X gate, solving the eigenvalue equation:

$$X\,|v\rangle = \lambda\,|v\rangle$$

Yields two eigenstates:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad \text{with eigenvalue} \quad +1 \Rightarrow X\,|+\rangle = +1 \cdot |+\rangle = |+\rangle \tag{13}$$

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \quad \text{with eigenvalue} \quad -1 \Rightarrow X\,|-\rangle = -1 \cdot |-\rangle = -\,|-\rangle \tag{14}$$

Example 2: Measurement in the X-basis

For example, if a qubit is in the state $|x\rangle$ and we measure it in the X-basis, the outcome is 1 with probability 1.

Proof. We want to show that if the qubit is in the state:

$$|x\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

And we measure it in the X-basis $\{|+\rangle, |-\rangle\}$, then the outcome is $+1$ with probability 1.
Measuring in the X-basis means using the eigenstates of the Pauli-X operator:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

These correspond to the new possible outcomes: $+1$ for $|+\rangle$ and -1 for $|-\rangle$.
If the system is in state $|\psi\rangle$, the probability of obtaining the outcome corresponding to basis state $|u\rangle$ is:

$$P(u) = |\langle u|\psi\rangle|^2$$

In our case, the system is in state $|\psi\rangle = |+\rangle$, so:

- Probability of outcome $+1$ (measuring $|+\rangle$):

$$P(+1) = |\langle +|+\rangle|^2$$

- Probability of outcome -1 (measuring $|-\rangle$):

$$P(-1) = |\langle -|+\rangle|^2$$

Since every normalized state has inner product 1 with itself, we have:

$$\langle +|+\rangle = 1$$

Therefore:

$$P(+) = |\langle +|+\rangle|^2 = |1|^2 = 1$$

So the probability of getting $+1$ is 1.
If we calculate the probability of getting -1, first write the bras and kets:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Changing the ket $|+\rangle$ to a bra:

$$\langle -| = \frac{1}{\sqrt{2}}(\langle 0| - \langle 1|)$$

Now we can compute the inner product:

$$
\begin{aligned}
\langle -|+\rangle &= \frac{1}{\sqrt{2}}\left(\langle 0| - \langle 1|\right)\frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right) \\
&\downarrow \text{ bring out the factors} \\
&= \frac{1}{2}\left(\langle 0| - \langle 1|\right)\left(|0\rangle + |1\rangle\right) \\
&\downarrow \text{ expand the product} \\
&= \frac{1}{2}\left(\langle 0|0\rangle + \langle 0|1\rangle - \langle 1|0\rangle - \langle 1|1\rangle\right) \\
&\downarrow \text{ use orthonormality } (\langle 0|0\rangle = \langle 1|1\rangle = 1,\ \langle 0|1\rangle = \langle 1|0\rangle = 0) \\
&= \frac{1}{2}\left(1 + 0 - 0 - 1\right) \\
&= 0
\end{aligned}
$$

Therefore:

$$P(-1) = |\langle -|+\rangle|^2 = |0|^2 = 0$$

So the probability of getting -1 is 0. QED

3.3.2.3 Pauli-Z (Phase Flip) Gate

The **Pauli-Z Gate**, also called the **Phase Flip Gate**, is one of the core **Pauli operators** used in quantum computing. It acts **only on the phase** of the qubit, **leaving the $|0\rangle$ amplitude unchanged** but **flipping the sign of the $|1\rangle$ amplitude**.

Matrix Representation

$$Z = \sigma_Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{15}$$

Action on a General Qubit

For a general state:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

Applying Z:

$$Z\,|\psi\rangle = aZ\,|0\rangle + bZ\,|1\rangle = a\,|0\rangle - b\,|1\rangle$$

As a result, Z **flips the sign of the $|1\rangle$ amplitude**, this is why it's called a **phase flip**. Finally, it's trivial to show that:

$$Z\,|0\rangle = |0\rangle \quad \text{(unchanged)}, \qquad Z\,|1\rangle = -\,|1\rangle \quad \text{(sign flip)}$$

So the **magnitudes** of the amplitudes remain unchanged and the **probabilities of measurement outcomes** in the standard basis are unaffected, but the **relative phase** between $|0\rangle$ and $|1\rangle$ is altered (since b changes to $-b$).

Example 3: Effect on Superposition States

The Hadamard (or X-basis) states are:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

These two states differ **only by their relative phase**:

- $|+\rangle$: relative phase 0.
- $|-\rangle$: relative phase π.

Applying Z on $|+\rangle$:

$$\begin{aligned} Z\,|+\rangle &= Z\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) \\ &= \frac{Z\,|0\rangle + Z\,|1\rangle}{\sqrt{2}} \\ &= \frac{|0\rangle - |1\rangle}{\sqrt{2}} \\ &= |-\rangle \end{aligned}$$

Applying Z on $|-\rangle$:

$$\begin{aligned} Z|-\rangle &= Z\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) \\ &= \frac{Z|0\rangle - Z|1\rangle}{\sqrt{2}} \\ &= \frac{|0\rangle + |1\rangle}{\sqrt{2}} \\ &= |+\rangle \end{aligned}$$

The Pauli-Z gate **does not change the probabilities** of measuring $|0\rangle$ or $|1\rangle$ in the computational basis. However, it **changes the relative phase**, transforming $|+\rangle$ into $|-\rangle$ and vice versa.

Geometric Interpretation: Rotation around the z-axis

On the Bloch sphere, the Pauli-Z gate performs a π **radians** (180°) **rotation around the z-axis**. In spherical coordinates:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

After Z rotation:

- The angle $\phi \to \phi + \pi$
- This phase shift **changes the sign of the complex amplitude for** $|1\rangle$

Since **measurement probabilities depend on** $|a|^2$ and $|b|^2$, which remain unchanged, the Z gate **does not affect measurement outcomes** in the standard basis. It only affects **interference and future gate interactions**.

Rotation Interpretation

From Equation 6 (page 46) and Equation 10 (page 46), a rotation around the z-axis by an angle θ is given by:

$$R_z(\theta) = \cos\left(\frac{\theta}{2}\right)I - i\sin\left(\frac{\theta}{2}\right)Z$$

For $\theta = \pi$, we obtain:

$$R_z(\pi) = -iZ$$

Therefore, the Pauli-Z gate corresponds to a **rotation of π around the z-axis, up to a global phase**:

$$Z \equiv R_z(\pi)$$

Since global phases are physically irrelevant, Z and $R_z(\pi)$ represent the same physical transformation.

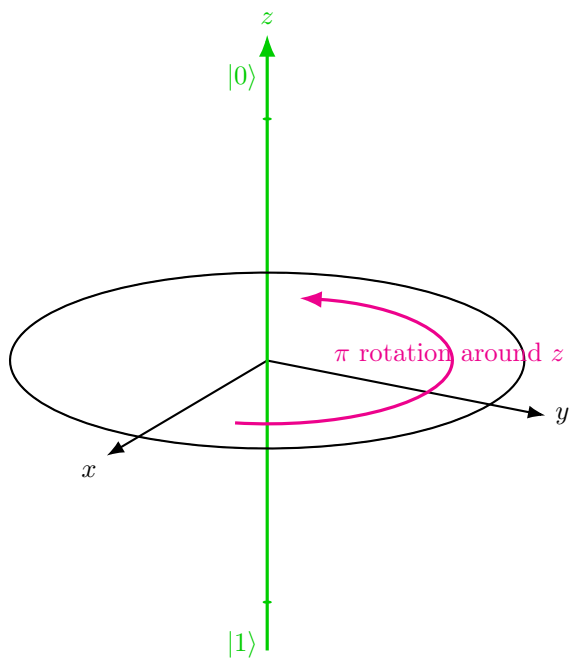


Figure 11: Eigenstates of the Pauli-Z gate. Since $|0\rangle$ and $|1\rangle$ lie on the z-axis, they are not moved by a rotation around that axis. The state $|1\rangle$ only acquires a global phase -1.

Key Properties

1. **Unitary**: $Z^\dagger = Z^{-1} = Z$. Ensuring normalization is preserved.
2. **Hermitian**: $Z^\dagger = Z$. Meaning it is equal to its own conjugate transpose.
3. **Self-Inverse**: $Z^{-1} = Z \Rightarrow Z^2 = I$. Applying the Pauli-Z gate twice returns the original state.
4. **Eigenvalues and Eigenstates**. The Pauli-Z operator has *eigenvalues* $+1$ and -1, with corresponding *eigenstates* $|0\rangle$ and $|1\rangle$:

   $$Z|0\rangle = +1 \cdot |0\rangle, \qquad Z|1\rangle = -1 \cdot |1\rangle$$

   This means that $|0\rangle$ and $|1\rangle$ are **unchanged in direction** by the action of Z (up to a possible phase factor). In particular:

   - $|0\rangle$ is completely unchanged.
   - $|1\rangle$ acquires only a phase factor $-1 = e^{i\pi}$.

   Therefore, these states are **invariant under the action of** Z (up to global phase), and represent the **natural states of the operator**.

Geometric Insight. On the Bloch sphere, the Pauli-Z gate corresponds to a rotation around the z-axis. The eigenstates $|0\rangle$ and $|1\rangle$ lie exactly on this axis (north and south poles). As a consequence, they are **not affected by the rotation**, which explains why they are eigenstates of Z.

In contrast, any state that is not aligned with the z-axis (for example $|+\rangle$, Figure 9) is rotated by the action of Z. This highlights the difference between:

- **Eigenstates**: unchanged (up to phase)
- **Superposition states**: geometrically rotated

3.3.2.4 Pauli-Y Gate

The **Pauli-Y Gate**, denoted Y, is one of the three fundamental **Pauli matrices** in quantum mechanics. It is less intuitive than Pauli-X and Z, but equally important, especially for its role in complex rotations and quantum interference.

Matrix Representation

$$Y = \sigma_Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Action on a General Qubit

At first glance, this looks similar to the Pauli-X gate because it exchanges $|0\rangle$ and $|1\rangle$. But there is something extra: X swaps the two basis states (i.e., $|0\rangle \leftrightarrow |1\rangle$), while Y swaps them **and** multiplies them by a phases of $\pm i$. So Pauli-Y is not just a bit flip (like Pauli-X), it is a bit flip together with a phase modification.

Given a qubit in state:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

Apply the Y gate:

$$Y\,|\psi\rangle = aY\,|0\rangle + bY\,|1\rangle = a\,(-i\,|1\rangle) + b\,(i\,|0\rangle) = ib\,|0\rangle - ia\,|1\rangle$$

So the **amplitudes are exchanged** (as with Pauli-X), but each one also acquires a **phase factor of i or $-i$**.

Geometric Interpretation: Rotation on the Bloch Sphere

The Pauli-Y gate corresponds to a **rotation around the y-axis by π radians** (180°). The effect on Bloch sphere coordinates:

- $\theta \to \pi - \theta$
- $\psi \to \pi - \psi$

This rotates the qubit across the y-axis, **altering both amplitudes and their phases**.

Rotation Interpretation

From Equation 6 (page 46) and Equation 10 (page 46), a rotation around the y-axis by an angle θ is given by:

$$R_y(\theta) = \cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) Y$$

For $\theta = \pi$, we obtain:

$$R_y(\pi) = -iY$$

Therefore, the Pauli-Y gate corresponds to a **rotation of π around the y-axis, up to a global phase**:

$$Y \equiv R_y(\pi)$$

Since global phases are physically irrelevant, Y and $R_y(\pi)$ represent the same physical transformation.

Pauli-Y as a Composite Gate

A very useful identity is:

$$Y = iXZ \tag{16}$$

This means that Pauli-Y can be understood as the combination of:

1. A Pauli-Z gate (phase flip);
2. Followed by a Pauli-X gate (bit flip);
3. Up to a global phase factor of i.

$$XZ = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

Now, multiply by i:

$$iXZ = i \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = Y$$

This identity confirms that **Pauli-Y can be viewed as a combination of Pauli-X and Pauli-Z with a global phase factor** i, which does not affect measurement outcomes.

Eigenstates of Y Gate

To find the eigenstates, we solve:

$$Y |v\rangle = \lambda |v\rangle$$

Where λ is the eigenvalue. The result is:

$$|i\rangle = \frac{1}{\sqrt{2}} (|0\rangle + i |1\rangle) \tag{17}$$

$$|-i\rangle = \frac{1}{\sqrt{2}} (|0\rangle - i |1\rangle) \tag{18}$$

With eigenvalues:

$$Y |i\rangle = +1 |i\rangle \qquad Y |-i\rangle = -1 |-i\rangle \tag{19}$$

Key Properties of Y Gate

1. **Unitary**: $Y^\dagger Y = I$. So it is a valid quantum gate that preserves normalization.
2. **Hermitian**: $Y^\dagger = Y$. So it is also an observable, meaning it corresponds to a measurable quantity in quantum mechanics.
3. **Self-inverse**: $Y^2 = I$. Applying the Y gate twice returns the qubit to its original state.

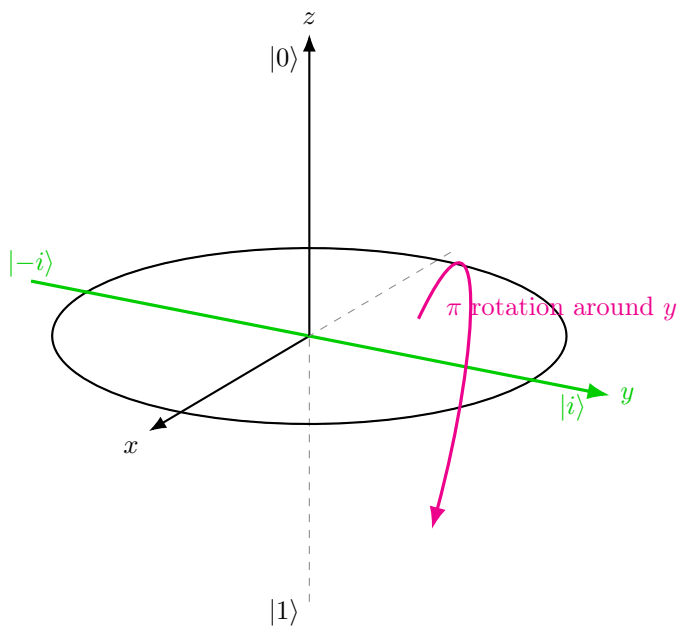


Figure 12: Eigenstates of the Pauli-Y gate. Since $|i\rangle$ and $|-i\rangle$ lie on the y-axis, they are not moved by a rotation around that axis. The state $|-i\rangle$ only acquires a global phase -1.

3.3.3 Phase Gate (S)

The **Phase Gate (S)** is a single-qubit gate that **adds a phase shift of $\frac{\pi}{2}$ ($90°$) to the amplitude of the $|1\rangle$ state**. It **leaves $|0\rangle$ unchanged**, like the Pauli-Z gate, but instead of a sign flip (π phase), it **introduces a complex phase factor** i.

Matrix Representation

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \tag{20}$$

Action on a General Qubit

For a state:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

Apply S:

$$S\,|\psi\rangle = a\,|0\rangle + ib\,|1\rangle$$

Only the $|1\rangle$ **amplitude gains a phase of** i, affecting interference but **not measurement probabilities** in the computational basis.

$$S\,|0\rangle = |0\rangle \quad \text{(unchanged)}, \qquad S\,|1\rangle = i\,|1\rangle \quad \left(\text{phase } + \frac{\pi}{2}\right)$$

Geometric Interpretation: Rotation around z-axis $\left(\frac{\pi}{2}\right)$

On the Bloch sphere, S performs a rotation of $\frac{\pi}{2}$ radians around the z-axis. This rotation **shifts the phase angle** $\phi \to \phi + \frac{\pi}{2}$, altering **how the qubit interferes** in later operations, especially when **Hadamard gates** are involved. In Figure 13, we see how S transforms the state $|+\rangle$ (on the x-axis) into $|i\rangle$ (on the y-axis) through a $\frac{\pi}{2}$ rotation around the z-axis, demonstrating the phase shift's effect on the qubit's state.

Relationship to Pauli-Z

S is often called the **square root of Z** because applying it twice yields the Pauli-Z gate: $S = \sqrt{Z}$. Pauli-Z is a **full phase flip**:

$$e^{i\pi} = -1$$

The S gate is a **partial phase shift**:

$$e^{i\frac{\pi}{2}} = i$$

So:

$$Z = e^{i\pi \cdot |1\rangle\langle 1|} \quad \text{and} \quad S = e^{i\frac{\pi}{2} \cdot |1\rangle\langle 1|}$$

Same transformation, but different phase angles. Let's verify $S^2 = Z$:

$$S \cdot S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & i^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z$$

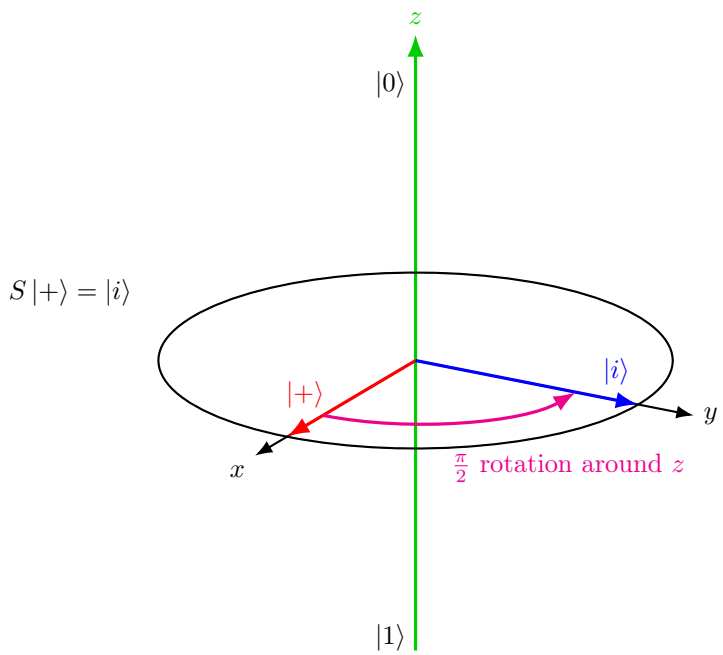


Figure 13: Geometric interpretation of the Phase gate S. The gate performs a rotation of $\frac{\pi}{2}$ around the z-axis, transforming $|+\rangle$ into $|i\rangle$.

Eigenvalues and Eigenstates

The Phase gate S is diagonal in the computational basis, so its eigenstates are $|0\rangle$ and $|1\rangle$:

$$S|0\rangle = 1 \cdot |0\rangle, \qquad S|1\rangle = i \cdot |1\rangle$$

Thus, the eigenvalues of S are:

$$\lambda_0 = 1, \qquad \lambda_1 = i$$

This means that $|0\rangle$ remains unchanged, while $|1\rangle$ acquires a phase factor $i = e^{i\frac{\pi}{2}}$.

Key Properties

1. **Unitary**: $S^\dagger = S^{-1}$, so normalization is preserved.
2. **Not Hermitian**: $S^\dagger \neq S$, therefore S is not an observable, it is only a transformation.
3. **Inverse**:
$$S^{-1} = S^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}$$
4. **Relation with Pauli-Z**: $S^2 = Z$, meaning that S is the square root of the Pauli-Z gate.
5. **Periodicity**: $S^4 = I$, since $i^4 = 1$.
6. **Phase Shift**: S applies a relative phase of $\frac{\pi}{2}$ to the $|1\rangle$ component.

3.3.4 Hadamard Gate (H)

The **Hadamard Gate (H)** **changes the basis representation of the qubit**. More concretely:

$$H : \{|0\rangle, |1\rangle\} \longleftrightarrow \{|+\rangle, |-\rangle\}$$

It maps computational basis states into equal superpositions and vice versa, allowing phase information to be transformed into measurable probabilities.

Matrix Representation

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \tag{21}$$

Action on Basis States

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle \tag{22}$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle \tag{23}$$

The $|+\rangle$ and $|-\rangle$ **states** are known as the **Hadamard basis**, or **superposition states**, with **equal probability amplitudes** for $|0\rangle$ and $|1\rangle$.

Superposition and Measurement

After applying H to $|0\rangle$, the resulting qubit is:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

This state is a **superposition** of $|0\rangle$ and $|1\rangle$, meaning that the qubit does not have a definite classical value prior to measurement. Instead, it is described by **probability amplitudes** associated with each basis state.

Upon measurement in the computational basis, the state collapses probabilistically:

- **Probability of 0**:

$$\left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$$

- **Probability of 1**:

$$\frac{1}{2}$$

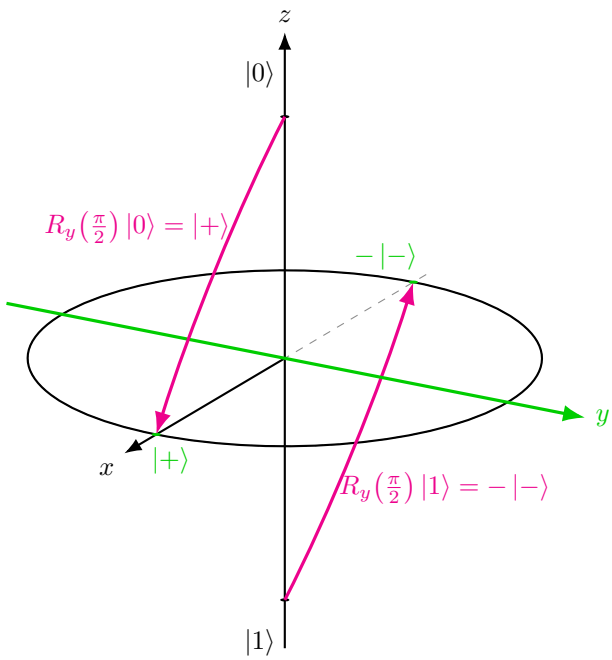
After the measurement:

- The qubit collapses to $|0\rangle$ with probability $\frac{1}{2}$, or to $|1\rangle$ with probability $\frac{1}{2}$.
- The original superposition is **destroyed**, and cannot be recovered.

Bloch Sphere Interpretation

On the Bloch sphere, the **Hadamard gate performs two sequential rotations**:

1. Rotation around the y-axis by $\frac{\pi}{2}$ radians (90°): $R_y\left(\frac{\pi}{2}\right)$.



The y-axis stays fixed. So the states on the y-axis $|i\rangle$ and $|-i\rangle$ do not move. Instead, the z-axis rotated toward the x-axis: $|0\rangle \to |+\rangle$ and $|1\rangle \to -|-\rangle$. So the first step already moves the computational basis from the z-axis to the x axis.

We can verify the matrix representation of $R_y\left(\frac{\pi}{2}\right)$ by using the exponential of the Pauli Y matrix (see page 46):

$$\begin{aligned} R_y(\theta) &= e^{-i\frac{\theta}{2}Y} \\ &= \cos\left(\frac{\theta}{2}\right) I - i\sin\left(\frac{\theta}{2}\right) Y \\ &= \cos\left(\frac{\theta}{2}\right)\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - i\sin\left(\frac{\theta}{2}\right)\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\ &= \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix} \end{aligned}$$

Now, substituting $\theta = \frac{\pi}{2}$ gives us the expected matrix for the first step of the Hadamard decomposition:

$$R_y\left(\frac{\pi}{2}\right) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

- Apply it to $|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}$:

$$\begin{aligned}
R_y\left(\frac{\pi}{2}\right)|0\rangle &= \frac{1}{\sqrt{2}}\begin{pmatrix}1 & -1\\1 & 1\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix}\\
&= \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}\\
&= \frac{1}{\sqrt{2}}\left(\begin{pmatrix}1\\0\end{pmatrix}+\begin{pmatrix}0\\1\end{pmatrix}\right)\\
&= \frac{1}{\sqrt{2}}\left(|0\rangle+|1\rangle\right)\\
&= \frac{|0\rangle+|1\rangle}{\sqrt{2}}\\
&= |+\rangle
\end{aligned}$$

  So $|0\rangle$, the north pole of the Bloch sphere, is rotated to the $+x$-axis, which is the $|+\rangle$ state.

- Apply it to $|1\rangle = \begin{pmatrix}0\\1\end{pmatrix}$:

$$\begin{aligned}
R_y\left(\frac{\pi}{2}\right)|1\rangle &= \frac{1}{\sqrt{2}}\begin{pmatrix}1 & -1\\1 & 1\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix}\\
&= \frac{1}{\sqrt{2}}\begin{pmatrix}-1\\1\end{pmatrix}\\
&= \frac{1}{\sqrt{2}}\left(\begin{pmatrix}0\\1\end{pmatrix}-\begin{pmatrix}1\\0\end{pmatrix}\right)\\
&= \frac{1}{\sqrt{2}}\left(|1\rangle-|0\rangle\right)\\
&= -\frac{|0\rangle-|1\rangle}{\sqrt{2}}\\
&= -|-\rangle
\end{aligned}$$

  This means that $|1\rangle$, the south pole of the Bloch sphere, moves to the $-x$-axis, but with a global phase of -1. Since global phases do not affect measurement outcomes, we can ignore it and say that $|1\rangle$ is effectively rotated to the $-x$-axis, which is the $|-\rangle$ state. In simple terms: $-|-\rangle \equiv |-\rangle$ up to a global phase.

2. Rotation around the x-axis by π radians (180°). This step is represented by the operator $R_x(\pi)$, which can be expressed as:

$$R_x(\pi) = e^{-i\frac{\pi}{2}X} = \cos\left(\frac{\pi}{2}\right)I - i\sin\left(\frac{\pi}{2}\right)X = -iX$$

   Applying $R_x(\pi)$ to the states on the x-axis ($|+\rangle$ and $|-\rangle$) does not change their position on the Bloch sphere, but it does introduce a phase factor

(which becomes a global phase only when applied to the entire state):

$$R_x(\pi)\,|+\rangle = -iX\,|+\rangle = -i\,|+\rangle$$
$$R_x(\pi)\,|-\rangle = -iX\,|-\rangle = -i(-\,|-\rangle) = i\,|-\rangle$$

Applying $R_x(\pi)$ to the superposition state $a\,|+\rangle - b\,|-\rangle$ (obtained from the first rotation) gives us:

$$\begin{aligned} a\,|+\rangle - b\,|-\rangle \\ R_x(\pi)(a\,|+\rangle - b\,|-\rangle) &= a(-i\,|+\rangle) - b(i\,|-\rangle) \\ &= -ia\,|+\rangle - ib\,|-\rangle \\ &= -i(a\,|+\rangle + b\,|-\rangle) \end{aligned}$$

Since global phases $-i$ do not affect measurement outcomes, we can ignore them and say that $|+\rangle$ and $|-\rangle$ remain on the x-axis. The second rotation does not change the position of the states on the Bloch sphere, but it is necessary to ensure that the transformation matches the Hadamard gate at the operator level, preserving the correct relative phase for all superpositions.

Why is the second rotation necessary? We **cannot** simply change $-\,|-\rangle$ to $|-\rangle$ inside a superposition, because that would change only one component and therefore change the **relative phase**. The fix comes from applying $R_x(\pi)$. After the first rotation, we have:

$$a\,|+\rangle - b\,|-\rangle$$

Now, applying the Pauli-X operator using the rotation $R_x(\pi)$ (page 46):

$$R_x(\pi) = -iX = -i\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{with} \quad X\,|+\rangle = |+\rangle\,, \quad X\,|-\rangle = -\,|-\rangle$$

Applying $R_x(\pi)$ to the superposition gives us:

$$\begin{aligned} R_x(\pi)(a\,|+\rangle - b\,|-\rangle) &= -iX(a\,|+\rangle - b\,|-\rangle) \\ &= -i\,(aX\,|+\rangle - bX\,|-\rangle) \\ &= -i\,(a\,|+\rangle - b(-\,|-\rangle)) \\ &= -i\,(a\,|+\rangle + b\,|-\rangle) \end{aligned}$$

Now the factor $-i$ multiplies the **entire state**, so it is a global phase:

$$-i(a\,|+\rangle + b\,|-\rangle) \equiv a\,|+\rangle + b\,|-\rangle$$

So the important point is:

$$a\,|+\rangle - b\,|-\rangle \not\equiv a\,|+\rangle + b\,|-\rangle$$

But:

$$R_x(\pi)(a\,|+\rangle - b\,|-\rangle) = -i(a\,|+\rangle + b\,|-\rangle) \equiv a\,|+\rangle + b\,|-\rangle$$

Insight. The second rotation **does not affect the geometric position** of the states, but **corrects the relative phase** structure, ensuring that the transformation is a valid unitary operator acting consistently on the entire Hilbert space.

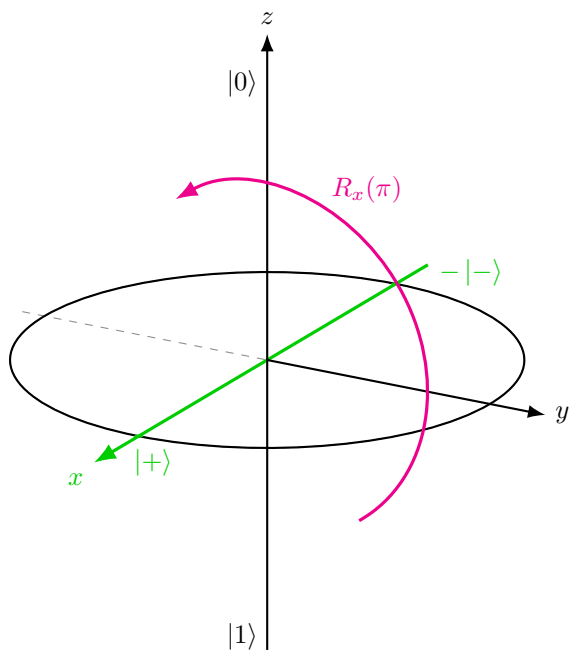


Figure 14: Second step of the Hadamard decomposition: rotation by π around the x-axis. States on the x-axis are not moved on the Bloch sphere; only global phases are introduced.

This brings $|0\rangle$ to the $+x$-axis ($|+\rangle$) and $|1\rangle$ to the $-x$-axis ($|-\rangle$). As a result, **qubit moves from pole to equator**, entering **maximum superposition**.

Eigenvalues and Eigenstates

The Hadamard operator has eigenvalues:

$$\lambda = +1, \qquad \lambda = -1$$

with corresponding eigenstates given by the solutions of:

$$H\,|v\rangle = \lambda\,|v\rangle$$

These eigenstates are superpositions of $|0\rangle$ and $|1\rangle$, and are aligned along an axis halfway between the x and z directions on the Bloch sphere. Precisely, the eigenvectors are:

$$|v_+\rangle = \cos\left(\frac{\pi}{8}\right)|0\rangle + \sin\left(\frac{\pi}{8}\right)|1\rangle$$
$$|v_-\rangle = \sin\left(\frac{\pi}{8}\right)|0\rangle - \cos\left(\frac{\pi}{8}\right)|1\rangle$$

Where $|v_+\rangle$ corresponds to the eigenvalue $+1$ and $|v_-\rangle$ corresponds to the eigenvalue -1; the cosine and sine factors ensure normalization of the eigenstates.

Unlike Pauli gates, the Hadamard eigenstates lie along a **diagonal axis between the computational basis z and the Hadamard basis x**, reflecting its role as a **basis transformation** rather than a simple bit or phase flip.

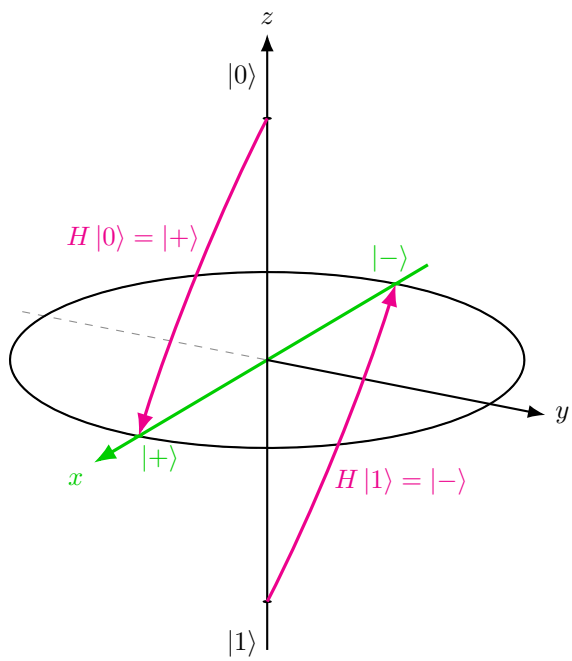


Figure 15: Bloch sphere interpretation of the Hadamard gate. The gate maps $|0\rangle$ to $|+\rangle$ and $|1\rangle$ to $|-\rangle$, moving computational basis states from the z-axis to the x-axis.

Key Properties

1. **Unitary**: $H^\dagger H = I$, so normalization is preserved.
2. **Hermitian**: $H^\dagger = H$, therefore it corresponds to an observable.
3. **Self-inverse**: $H^2 = I$. Applying the Hadamard gate twice returns the original state.
4. **Basis transformation**: H maps the computational basis $\{|0\rangle, |1\rangle\}$ to the Hadamard basis $\{|+\rangle, |-\rangle\}$:

$$H|0\rangle = |+\rangle, \qquad H|1\rangle = |-\rangle$$

5. **Interference enabler**: H converts phase differences into amplitude differences, making them observable through measurement.
6. **Axis mixing**: H transforms Pauli operators as:

$$HZH = X, \qquad HXH = Z$$

3.4 Properties

Single-qubit gates exhibit a rich set of mathematical and geometric properties, all of which stem from their nature as unitary transformations on a two-level quantum system. Understanding these properties is key to **predicting gate behavior**, **designing quantum circuits**, and **analyzing quantum algorithms**.

1. All Quantum Gates Are Equivalent to Rotations

A profound insight in quantum computing is that **every single-qubit unitary gate corresponds to a rotation of the qubit's state vector on the Bloch sphere**.

- **Bloch Sphere** is a geometric representation of a qubit's state, where any point on the sphere represents a valid pure state.
- A **unitary operation** on qubit results in **rotating the vector** on this sphere **without changing its length** (preserving probability).

Some examples of rotations:

- `X Gate` → Rotation π around x-axis (section 3.3.2.2, page 47)
- `Z Gate` → Rotation π around z-axis (section 3.3.2.3, page 52)
- `Y Gate` → Rotation π around y-axis (section 3.3.2.4, page 56)
- `S Gate` → Rotation $\frac{\pi}{2}$ around z-axis (section 3.3.3, page 59)
- `H Gate` → Rotation $\frac{\pi}{2}$ around y, then π around x (section 3.3.4, page 61)

Why is this important? Quantum computation is fundamentally about **state rotations**, understanding gates as rotations helps visualize **interference**, **phase shifts**, and **entanglement creation**.

2. Reversibility: Applying a Gate Twice

Many single-qubit gates, especially the Pauli gates, exhibit the property of **involution**: applying the same gate twice **returns the qubit to its original state**. Mathematically it can be expressed as:

$$X^2 = Y^2 = Z^2 = I$$

This is due to the fact that **two** 180° **rotations around the same axis bring the vector back** to its original position on the Bloch sphere.

Proof that X Gate has involution.

$$X^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

These identities confirm that **Pauli gates are their own inverses**, and **all unitary gates are reversible**. QED

3. Global Phases Do Not Matter

Quantum gates may **introduce a global phase** (a constant complex factor $e^{i\phi}$) to a state. Physically, **global phases are unobservable**, meaning they do **not affect measurement probabilities**. For example:

- States $|\psi\rangle$ and $e^{i\phi}|\psi\rangle$ are physically indistinguishable.
- The Y gate $= iXZ$, where i is a global phase that can be ignored in practice.

4. Composite Gates and Commutativity

- Gates can be **combined** by **matrix multiplication** (note: non commutative in general).
- **Order matters**: $AB \neq BA$ for most gates.
- Exception: **Gate Z and Gate S** are **commuted** because they are both **diagonal phase gates**.

Property	Pauli Gates (X, Y, Z)	Phase Gate (S)	Hadamard (H)
Rotation Axis	x, y, z (π radians)	z $\left(\frac{\pi}{2}\text{radians}\right)$	y $\left(\frac{\pi}{2}\right)$ + x (π)
Involution $G^2 = I$	✔	✘ ($S^2 = Z$)	✔
Self-adjoint $G = G^\dagger$	✔	✘	✔
Unitary	✔	✔	✔
Phase Introduction	Z, S	i for $\|1\rangle$	✔ (interference)
Creates Superposition?	✘ (X, Y, Z alone)	✘	✔

Table 2: Summary of Properties.

3.5 When Does a Gate Create Superposition?

In this section, we will answer the question: *when does a quantum gate create superposition?* More precisely, if we start from a **basis state** $|0\rangle$ or $|1\rangle$, when does a gate produce:

$$a\,|0\rangle + b\,|1\rangle \quad \text{with} \quad a, b \neq 0\,?$$

What does "*create superposition*" really mean? It means that the output must contain **both basis states**. So:

- $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is not a superposition, it's just $|0\rangle$.
- $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is not a superposition, it's just $|1\rangle$.
- $\begin{pmatrix} a \\ b \end{pmatrix}$ with $a, b \neq 0$ is a superposition, because it contains both $|0\rangle$ and $|1\rangle$ components. Indeed, we can write it as:

$$a\,|0\rangle + b\,|1\rangle = a\begin{pmatrix} 1 \\ 0 \end{pmatrix} + b\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$$

How does a matrix act on a basis state?

The main idea is that each column of the matrix tells us where a basis vector goes. For example, for a matrix:

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

The first column $\begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix}$ tells us where $|0\rangle$ goes, and the second column $\begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix}$ tells us where $|1\rangle$ goes. In general, a gate creates superposition if one of its columns has **two non-zero entries**.

Constraint: Unitarity

Not every matrix can be a valid quantum gate. Quantum gates must be **unitary**, meaning:

$$A^\dagger A = I$$

This imposes strong constraints on the matrix entries. In particular, the columns must be:

- **Normalized**, total probability is 1: $\|\psi\|^2 = |a|^2 + |b|^2 = 1$
- **Orthogonal**, inner product is zero: $\langle v|w\rangle = 0$

As a consequence, some matrices that appear to create superposition are **not physically allowed**, because they are not unitary.

Key Insight

Superposition arises from the **mixing of basis states**. Therefore:

- A gate **creates superposition** if at least one of its columns contains **two non-zero entries**.
- A gate **cannot create superposition** if its columns contain only one non-zero entry (as in diagonal or triangular matrices).

In particular:

- Gates like H (Hadamard) create superposition because they mix $|0\rangle$ and $|1\rangle$.
- Gates like X, Y, Z or S do not create superposition because they only modify phases and do not mix basis states.

Proof that A cannot create superposition. Consider the gate:

$$A = \begin{pmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{pmatrix}$$

To create superposition from a basis state, at least one column of A must contain two non-zero entries.

Applying A to $|0\rangle$ gives:

$$A|0\rangle = \begin{pmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a_{11} \cdot 1 + a_{12} \cdot 0 \\ 0 \cdot 1 + a_{22} \cdot 0 \end{pmatrix} = \begin{pmatrix} a_{11} \\ 0 \end{pmatrix} = a_{11}|0\rangle$$

Therefore, $|0\rangle$ is not mapped to a superposition because the second entry is zero.

Applying A to $|1\rangle$ gives:

$$A|1\rangle = \begin{pmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a_{11} \cdot 0 + a_{12} \cdot 1 \\ 0 \cdot 0 + a_{22} \cdot 1 \end{pmatrix} = \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix} = a_{12}|0\rangle + a_{22}|1\rangle$$

This would be a superposition only if both $a_{12} \neq 0$ and $a_{22} \neq 0$.

However, since A is a quantum gate, it must be unitary (property of quantum gates):

$$A^\dagger A = I$$

Assuming A is real, we have $A^\dagger = A^T$, so:

$$A^T = \begin{pmatrix} a_{11} & 0 \\ a_{12} & a_{22} \end{pmatrix}$$

Hence:

$$A^T A = \begin{pmatrix} a_{11} & 0 \\ a_{12} & a_{22} \end{pmatrix} \cdot \begin{pmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{pmatrix} = \begin{pmatrix} a_{11}^2 & a_{11}a_{12} \\ a_{11}a_{12} & a_{12}^2 + a_{22}^2 \end{pmatrix}$$

Since A is unitary (i.e., $A^\dagger A = I$), this matrix must equal the identity:

$$\begin{pmatrix} a_{11}^2 & a_{11}a_{12} \\ a_{11}a_{12} & a_{12}^2 + a_{22}^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Therefore, to satisfy this equality, the following equations must hold:

$$\begin{cases} a_{11}^2 = 1 \\ a_{11}a_{12} = 0 \\ a_{12}^2 + a_{22}^2 = 1 \end{cases}$$

From $a_{11}^2 = 1$, we get $a_{11} = \pm 1$, so $a_{11} \neq 0$. Therefore, from $a_{11}a_{12} = 0$, it follows that:

$$a_{12} = 0$$

Substituting this into the last equation gives:

$$a_{22}^2 = 1 \quad \Rightarrow \quad a_{22} = \pm 1$$

Hence the matrix becomes diagonal:

$$A = \begin{pmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{pmatrix}$$

Since A is diagonal, it only modifies the phases of $|0\rangle$ and $|1\rangle$, without mixing them. Thus, when we apply A to $|0\rangle$ and $|1\rangle$, we get:

$$A\,|0\rangle = \pm\,|0\rangle\,, \qquad A\,|1\rangle = \pm\,|1\rangle$$

So neither $|0\rangle$ nor $|1\rangle$ is mapped to a superposition, because the output states are just $|0\rangle$ or $|1\rangle$ (up to a global phase). Thus, A cannot create superposition (i.e., it cannot produce a state with both $|0\rangle$ and $|1\rangle$ components). QED

3.6 Single-Qubit Quantum Circuits

A **Quantum Circuit** is a **mathematical model** for quantum computation, **composed of sequential operations applied to qubits**. In other words, a quantum circuit is a **model of computation** that describes **how a quantum state evolves step by step**. The operations in a quantum circuit include initializations, unitary gates, and measurements. In this section, we focus on circuits involving a single qubit, which form the foundation for understanding multi-qubit systems.

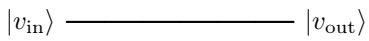
Circuit Elements

A quantum circuit typically consists of three main elements:

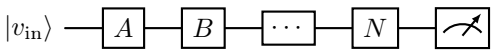
1. **Initialization** (initial state)
   - Qubit is prepared in a **known state**, typically $|0\rangle$ or $|1\rangle$.
   - Essential **starting point** for any computation.
2. **Gates**
   - Single-qubit **unitary operations** (e.g., X, Z, H, S).
   - Transform the qubit's state **reversibly**.
3. **Measurement**
   - **Irreversible process**.
   - Collapses the qubit to $|0\rangle$ or $|1\rangle$ with probabilities dictated by the quantum state's amplitudes.

Quantum Circuit Diagram: Visual Language

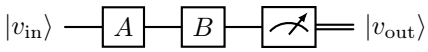
- **Horizontal lines** represent the **timeline of a single qubit**.



- **Boxes** on the line represent **gates** (letters) or **measurements** (tachometer icon).



- **Double lines** (if present) represent **classical information** (e.g., measurement result).



Left to right: The **circuit evolves over time from left (input) to right (output)**.

Gate Sequence: Serial Gates

Consider two gates A and B applied in sequence:

$$|v_{\text{in}}\rangle \;—\boxed{A}—\boxed{B}—\; |v_{\text{out}}\rangle$$

1. First apply A, then B.
2. **Attention**: the **order of matrix multiplication** is **reversed** when we write the combined effect of the gates as a single matrix. The gate that is applied **last** (B) appears **first** in the product.

   $$v_{\text{out}} = BAv_{\text{in}}$$

   Why? Because first apply A to the input state v_{in}, then apply B to the result of Av_{in}.
3. The **combined effect** of the two gates is equivalent to a single gate C = BA:

   $$|v_{\text{in}}\rangle \;—\boxed{BA}—\; |v_{\text{out}}\rangle \quad \equiv \quad |v_{\text{in}}\rangle \;—\boxed{A}—\boxed{B}—\; |v_{\text{out}}\rangle$$

   $$v_{\text{out}} = Cv_{\text{in}} = BAv_{\text{in}}$$

   Obviously, if we merge the gates into a single gate C, we must respect the order of operations, which is why $C = BA$ and not AB.

Example 4: Hadamard followed by Hadamard

Let's compute H followed by H to $|0\rangle$:

$$|0\rangle \;—\boxed{H}—\boxed{H}—\; |v_{\text{out}}\rangle$$

1. Apply H:

$$\begin{aligned} H\,|0\rangle &= \frac{1}{\sqrt{2}} \cdot \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \cdot \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \\ &= \frac{1}{\sqrt{2}} \left(|0\rangle + |1\rangle \right) = |+\rangle \end{aligned}$$

2. Applying H again, we **restore the original state due to the involution properties** ($H^2 = I$):

$$H\left(H\,|0\rangle\right) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$

3.7 Outer Product of Kets

In quantum mechanics, **Bra-Ket notation** (introduced by Dirac) is a concise and powerful tool for representing quantum states and operations. It uses **kets** $|\psi\rangle$ for **vectors** (states), and **bras** $\langle\psi|$ for their **Hermitian adjoints** (dual vectors).

Given two kets:

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \qquad |b\rangle = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

The **Outer Product** $|a\rangle\langle b|$ is a **matrix**, formed by:

$$|a\rangle\langle b| = |a\rangle \otimes \langle b| = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \begin{pmatrix} \overline{b_1} & \overline{b_2} \end{pmatrix} = \begin{pmatrix} a_1\overline{b_1} & a_1\overline{b_2} \\ a_2\overline{b_1} & a_2\overline{b_2} \end{pmatrix} \tag{24}$$

Where the symbol $\otimes$ is the **tensor product operator**. This operation **maps a vector to a matrix**.

Why is the outer product useful? Because **all quantum gates can be written using outer products of kets**. For example, the **Pauli-X gate** can be expressed as:

$$X = |0\rangle\langle 1| + |1\rangle\langle 0| = \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix}}_{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix}}_{\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

The **Inner Product** $\langle b|a\rangle$ is a scalar:

$$\langle b|a\rangle = \overline{b_1}a_1 + \overline{b_2}a_2 \tag{25}$$

So:

- **Inner product** $\rightarrow$ **projection**, result is number (scalar).
- **Outer product** $\rightarrow$ **matrix**, result is operator.

Associativity between outer and inner products

A key property of the outer product is how it interacts with inner products. Specifically:

$$(|a\rangle\langle b|)\,|c\rangle = |a\rangle\,(\langle b|c\rangle) \tag{26}$$

This identity shows that the outer product acts as an operator: it first computes the inner product $\langle b|c\rangle$ (a scalar), and then scales the vector $|a\rangle$.

Proof the Associativity. In this proof, we will demonstrate that applying the outer product $|a\rangle\langle b|$ to a ket $|c\rangle$ is equivalent to scaling $|a\rangle$ by the inner product $\langle b|c\rangle$.

Let's define:

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \qquad |b\rangle = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \qquad |c\rangle = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

1. Compute **inner product** $\langle b|c\rangle$:

$$\langle b|c\rangle = \overline{b_1}c_1 + \overline{b_2}c_2 = \alpha$$

   Where α is a **scalar** value.

2. Multiply **scalar** α with $|a\rangle$ (vector):

$$|a\rangle\,\alpha = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \alpha = \begin{pmatrix} a_1 \cdot \alpha \\ a_2 \cdot \alpha \end{pmatrix} = \begin{pmatrix} a_1 \cdot \left(\overline{b_1}c_1 + \overline{b_2}c_2\right) \\ a_2 \cdot \left(\overline{b_1}c_1 + \overline{b_2}c_2\right) \end{pmatrix}$$

3. Compute outer product $|a\rangle\langle b|$ (matrix):

$$|a\rangle\,\langle b| = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \begin{pmatrix} \overline{b_1} & \overline{b_2} \end{pmatrix} = \begin{pmatrix} a_1\overline{b_1} & a_1\overline{b_2} \\ a_2\overline{b_1} & a_2\overline{b_2} \end{pmatrix}$$

4. Multiply $|a\rangle\langle b|$ by $|c\rangle$:

$$\left(|a\rangle\,\langle b|\right)|c\rangle = \begin{pmatrix} a_1\overline{b_1} & a_1\overline{b_2} \\ a_2\overline{b_1} & a_2\overline{b_2} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} a_1 \cdot \left(\overline{b_1}c_1 + \overline{b_2}c_2\right) \\ a_2 \cdot \left(\overline{b_1}c_1 + \overline{b_2}c_2\right) \end{pmatrix}$$

The second and fourth steps yield the same result. This proves that **outer product applied to a ket is equivalent to inner product $\langle b|c\rangle$ scaled by $|a\rangle$.** QED

3.8 Measurement

In quantum computing, **measurement** is a special type of operation that differs fundamentally from unitary gates. While gates perform reversible transformations, **measurement is irreversible**. It allows us to **gain information about the state of a qubit**, but at the price of **collapsing the quantum state**.

Measurement Operator: Matrix Form

A **measurement operator** is a **non-unitary**, **non-invertible** matrix. For a measurement **along a specific direction**, represented by a ket $|k\rangle$, the associated **projector** is (outer product):

$$M_k = |k\rangle\langle k| \tag{27}$$

This is called a **Projection Operator**, it projects any vector **onto the direction of** $|k\rangle$.

Effect of Measurement

Given a qubit $|\psi\rangle$, applying M_k yields the **(unnormalized) projected vector**:

$$|\psi_k\rangle = M_k\,|\psi\rangle = |k\rangle\,\langle k|\psi\rangle = (|k\rangle\,\langle k|)\,|\psi\rangle \tag{28}$$

To obtain the **new normalized state**, we **divide by the square root of the probability**:

$$|\psi_k^{\text{norm}}\rangle = \frac{M_k \cdot |\psi\rangle}{\sqrt{\langle\psi\,|M_k|\,\psi\rangle}} = \frac{\langle k|\psi\rangle \cdot |k\rangle}{\sqrt{|\langle k|\psi\rangle|^2}} = |k\rangle \tag{29}$$

So after measurement, the qubit **collapses** to $|k\rangle$, the **eigenstate corresponding to the measurement outcome**.

Why Normalization is Needed

After applying the measurement operator $M_k = |k\rangle\,\langle k|$ (Equation 27) to a state $|\psi\rangle$ (Equation 28), we obtain:

$$|\psi_k\rangle = M_k\,|\psi\rangle = |k\rangle\,\langle k|\psi\rangle$$

This vector is generally **not normalized**. Its norm is:

$$\|\,|\psi_k\rangle\,\|^2 = \langle\psi|\,M_k\,|\psi\rangle = |\,\langle k|\psi\rangle\,|^2 = p_k$$

Where p_k is the **probability of obtaining outcome** k.

Deepening: Why $\|\,|\psi_k\rangle\,\|^2 = p_k$?

We want to show:

$$\||\psi_k\rangle\|^2 = \langle\psi|\,M_k\,|\psi\rangle = |\langle k|\psi\rangle|^2 = p_k$$

1. **Definition of squared norm**. The squared norm of a ket is:

$$\||\psi_k\rangle\|^2 = \langle\psi_k|\psi_k\rangle$$

Since $|\psi_k\rangle = M_k|\psi\rangle$, we can write:

$$\||\psi_k\rangle\|^2 = \underbrace{(M_k|\psi\rangle)^\dagger}_{\langle\psi_k|}\underbrace{(M_k|\psi\rangle)}_{|\psi_k\rangle}$$

We have only expanded the definition of the squared norm and substituted $|\psi_k\rangle$.

2. **Take the dagger †**. The dagger of a product of operators is the product of the daggers in reverse order:

$$(AB)^\dagger = B^\dagger A^\dagger$$

Applying this to $(M_k|\psi\rangle)^\dagger$:

$$(M_k|\psi\rangle)^\dagger = \langle\psi| M_k^\dagger$$

And substituting back into the expression for the squared norm:

$$\||\psi_k\rangle\|^2 = \langle\psi| M_k^\dagger M_k|\psi\rangle$$

3. **Use the Hermitian property of** M_k (explained later page 79). Since M_k is Hermitian, we have $M_k^\dagger = M_k$. Therefore:

$$\||\psi_k\rangle\|^2 = \langle\psi| M_k M_k|\psi\rangle$$

4. **Use the idempotent property of** M_k (explained later page 79). Since M_k is idempotent, we have $M_k^2 = M_k$. Therefore:

$$\||\psi_k\rangle\|^2 = \langle\psi| M_k|\psi\rangle$$

5. **Substitute the definition of** M_k. Since $M_k = |k\rangle\langle k|$, we can write:

$$\||\psi_k\rangle\|^2 = \langle\psi|(|k\rangle\langle k|)|\psi\rangle = \langle\psi|k\rangle\langle k|\psi\rangle$$

6. **Use the property of inner products**. The inner product $\langle\psi|k\rangle$ is the complex conjugate of $\langle k|\psi\rangle$:

$$\langle\psi|k\rangle = \overline{\langle k|\psi\rangle}$$

Therefore (page 7):

$$\||\psi_k\rangle\|^2 = \overline{\langle k|\psi\rangle}\langle k|\psi\rangle = |\langle k|\psi\rangle|^2 = p_k$$

This is exactly the probability p_k of obtaining outcome k when measuring the state $|\psi\rangle$ with the measurement operator M_k.

Important. The result of a measurement must be a valid quantum state, and therefore it must be **normalized** (i.e., have unit norm).

To restore normalization, we divide the projected vector by its norm:

$$|\psi_k^{\text{norm}}\rangle = \frac{|\psi_k\rangle}{\| \, |\psi_k\rangle \, \|} = \frac{|k\rangle \langle k|\psi\rangle}{\sqrt{|\langle k|\psi\rangle|^2}}$$

Observe that:

$$\frac{\langle k|\psi\rangle}{|\langle k|\psi\rangle|}$$

Is a complex number of magnitude 1, i.e., a **global phase**. Since global phases do not affect physical states, we conclude:

$$|\psi_k^{\text{norm}}\rangle = |k\rangle$$

Interpretation Measurement can be understood as a two-step process:

1. **Projection**: extract the component of $|\psi\rangle$ along $|k\rangle$ (this determines the probability p_k).
2. **Renormalization**: rescale the result to obtain a valid quantum state.

Probability of Outcome

The **probability of measuring the qubit in the state** $|k\rangle$ is:

$$p_k = \langle\psi|M_k|\psi\rangle = |\langle k|\psi\rangle|^2 = \left|\overline{k_1} \cdot \psi_1 + \overline{k_2} \cdot \psi_2\right|^2 \tag{30}$$

This is the **Born rule**, a fundamental postulate of quantum mechanics. In quantum computing we have:

$$\begin{aligned} p_0 &= \langle\psi|M_0|\psi\rangle = |\langle 0|\psi\rangle|^2 = |1 \cdot \psi_1 + \cancel{0 \cdot \psi_2}| \\ p_1 &= \langle\psi|M_1|\psi\rangle = |\langle 1|\psi\rangle|^2 = |\cancel{0 \cdot \psi_1} + 1 \cdot \psi_2| \end{aligned}$$

Remark: $\langle\psi|M_k|\psi\rangle$

This is called a quadratic form in linear algebra, and in quantum mechanics it represents the **probability of the measurement outcome** k when **measuring the qubit** $|\psi\rangle$ with **measurement operator** M_k. Suppose a qubit ψ:

$$|\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix} \qquad \text{with } |a|^2 + |b|^2 = 1 \text{ (normalization condition)}$$

If we apply the measurement operator M_0:

$$M_0|\psi\rangle = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a \\ 0 \end{pmatrix}$$

And we finally calculate the bra $\langle\psi|$:

$$\langle\psi| \begin{pmatrix} a \\ 0 \end{pmatrix} = \begin{pmatrix} \overline{a} & \overline{b} \end{pmatrix} \begin{pmatrix} a \\ 0 \end{pmatrix} = \overline{a} \cdot a = |a|^2$$

In other words, this is the probability of measuring 0: p_0.

Standard Basis Measurement

In quantum computing, we usually measure in the **computational basis** $\{|0\rangle, |1\rangle\}$. The **projectors** are:

$$\begin{aligned} M_0 &= |0\rangle\langle 0| &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \\ M_1 &= |1\rangle\langle 1| &= \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned} \tag{31}$$

For a **general qubit** $|\psi\rangle = a|0\rangle + b|1\rangle$, the **probabilities** are:

$$p_0 = |a|^2 \qquad p_1 = |b|^2 \tag{32}$$

After measurement, the qubit **collapses to** $|0\rangle$ **with probability** $|a|^2$, or to $|1\rangle$ **with probability** $|b|^2$.

Key Properties of Measurement Operator

- **Hermitian**: $M_k^\dagger = M_k$, which means that M_k **is equal to its conjugate transpose**. This implies that M_k has real eigenvalues and orthogonal eigenvectors.

$$\begin{aligned} M_k^\dagger &= (|k\rangle\langle k|)^\dagger \\ &\downarrow \text{ use the rule } (AB)^\dagger = B^\dagger A^\dagger \\ &= (\underbrace{|k\rangle}_{A}\underbrace{\langle k|}_{B})^\dagger \\ &= (\langle k|)^\dagger (|k\rangle)^\dagger \\ &= |k\rangle\langle k| \\ &= M_k \end{aligned}$$

- **Idempotent**: once applied, **applying it again does nothing**:

$$M_k^2 = M_k$$

  A simple proof:

$$M_k^2 = |k\rangle |k\rangle\langle k| \langle k| = |k\rangle\langle k| = M_k$$

  This reflects that **after projection, the qubit is already in the state** $|k\rangle$.

- **Non-unitary** and **Non-reversible**: measurement **destroys superposition** and **cannot be undone**.

4 Multiple Qubit Gates

4.1 Multiple Qubit States

Now that we have a solid understanding of single qubits and their states, we can transition to discussing many qubits as one joint system, rather than just one vector with one qubit.

With **one qubit**, life is simple because the state of the system can be described by a single vector in a two-dimensional complex vector space:

$$|v\rangle = a\,|0\rangle + b\,|1\rangle$$

With **two qubits**, something radical happens. We can no longer describe the system by looking at each qubit separately. Instead, we must describe **one global state** that captures the behavior of both qubits together. For example:

$$|v\rangle = c_0\,|00\rangle + c_1\,|01\rangle + c_2\,|10\rangle + c_3\,|11\rangle$$

This is not just a simple combination of two single-qubit states.

Exponential Explosion. The first thing to notice is that the number of amplitudes needed to describe the state of a system grows exponentially with the number of qubits.

- For 1 qubit, we need 2 amplitudes: a and b.
- For 2 qubits, we need 4 amplitudes: c_0, c_1, c_2, and c_3.
- For 3 qubits, we need 8 amplitudes: d_0, d_1, d_2, d_3, d_4, d_5, d_6, and d_7.
- For n qubits, we need 2^n amplitudes to fully describe the state of the system.

This is why quantum computing is so powerful: it can **process many possibilities at the same time** and **manipulate them so that only the correct ones survive**. This will become clear when we introduce entanglement (page 89).

The Key New Phenomenon: Entanglement

Entanglement is the key new phenomenon when dealing with multiple qubits. It refers to quantum states that cannot be expressed as a tensor product of individual qubit states. For example, the state:

$$\frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right)$$

Is an entangled state. In such cases, the system must be described globally, and it is not possible to assign independent quantum states to each qubit.

If one qubit is measured, the outcome is random, but it determines the outcome of the other qubit due to the structure of the global state. This leads to strong

correlations between measurements results. However, this does not imply any form of information transfer between the qubits.

Entanglement is a fundamental resource in quantum computing, enabling correlations that have no classical counterpart, especially when combined with interference effects in quantum algorithms.

- Entanglement does not mean that one qubit sends information to the other. The observed correlations arise from the collapse of a **global quantum state**, not from communication between subsystems.
- Before measurement, individual qubits in an entangled state do **not have a well-defined state**. Only the joint system is well-defined.
- Each qubit individually appears **random**, but measurement outcomes are **correlated**.
- Entanglement alone is not the source of quantum computational power; it becomes powerful when combined with **interference** and quantum operations.

New tools we need

To handle this new world of multiple qubits, we need new mathematical tools. We will introduce the **tensor product**, which allows us to construct the joint state of multiple qubits; we will also introduce the **multi-qubit gates**, which are operations that can manipulate the joint state of multiple qubits simultaneously; and we will discuss how to **measure** multiple qubits and understand the resulting probabilities.

4.1.1 Tensor Product

Let's start with something simple. As in previous sections, we have a simple one-qubit state. Now, we want to **add a second qubit to the system** to create a multi-qubit state. For simplicity, we will use only two qubits. *How do we describe the **combined system**?* Simply keeping them separate is not enough because we would miss correlations between the two qubits. We need a way to combine the two qubits into a **single object**. The tool for doing so is the **tensor product**.

Example 1: Dice Analogy

Imagine we have two dice. Each die can be in one of six states (1, 2, 3, 4, 5, or 6). If we want to describe the combined state of both dice, we can't just list their individual states separately. Instead, we need to consider all possible combinations of their states. The tensor product allows us to create a new space that includes all these combinations, giving us a complete description of the system.

The **Tensor Product** is a **mathematical operation** that takes two vectors (or more generally, two mathematical objects):

$$|v_A\rangle = a_0 |0\rangle + a_1 |1\rangle \quad |v_B\rangle = b_0 |0\rangle + b_1 |1\rangle$$

And produces a new vector that represents the combined state of the two systems:

$$\begin{aligned} |v_A\rangle \otimes |v_B\rangle &= (a_0 |0\rangle + a_1 |1\rangle) \otimes (b_0 |0\rangle + b_1 |1\rangle) \\ &= a_0 b_0 |00\rangle + a_0 b_1 |01\rangle + a_1 b_0 |10\rangle + a_1 b_1 |11\rangle \end{aligned} \tag{33}$$

Here, the resulting state $|v_A\rangle \otimes |v_B\rangle$ is a **superposition of all possible combinations of the states of the two qubits**.

? Why amplitudes multiply? For example, why is the amplitude of $|00\rangle$ equal to $a_0 b_0$? A qubit is a vector:

$$|v_A\rangle = \begin{pmatrix} a_0 \\ a_1 \end{pmatrix}, \quad |v_B\rangle = \begin{pmatrix} b_0 \\ b_1 \end{pmatrix}$$

The tensor product is **defined** as:

$$|v_A\rangle \otimes |v_B\rangle = \begin{pmatrix} a_0 b_0 \\ a_0 b_1 \\ a_1 b_0 \\ a_1 b_1 \end{pmatrix} \tag{34}$$

So amplitudes multiply because **that's how the tensor product is defined**. *But why is it defined like that?* Because we want a rule that satisfies two requirements:

1. **Linearity**. Quantum mechanics is linear:

$$(a_0 |0\rangle + a_1 |1\rangle) \otimes |v_B\rangle = a_0 (|0\rangle \otimes |v_B\rangle) + a_1 (|1\rangle \otimes |v_B\rangle)$$

The tensor product must be **bilinear**, i.e., linear in each argument, so that superpositions are preserved when combining quantum states.

2. **Basis consistency**:

$$|0\rangle \otimes |0\rangle = |00\rangle\,, \quad |0\rangle \otimes |1\rangle = |01\rangle\,, \quad |1\rangle \otimes |0\rangle = |10\rangle\,, \quad |1\rangle \otimes |1\rangle = |11\rangle$$

The tensor product must map basis states of individual systems to a consistent basis of the joint system (e.g., $|i\rangle \otimes |j\rangle = |ij\rangle$), preserving the structure of the state space.

This structure ensures that probabilities factorize correctly for independent systems.

⚠ Not all states can be factored

Given two qubits, their joint state lives in a 4-dimensional vector space. However, **not all states** in this 4-dimensional space can be written as a tensor product of two single-qubit states. Precisely, if a multi-qubit state can be written as a **tensor product**, then it is **separable**, and we can recover the individual qubit states. But if it is **not a tensor product**, then it is **entangled**, and we cannot assign well-defined quantum states to each qubit individually.

In general, exists a **necessary (and sufficient) condition** for being a tensor product for two qubits. Given a 2-qubit state:

$$|\psi\rangle = (c_0, c_1, c_2, c_3)$$

Where c_0, c_1, c_2, c_3 are the amplitudes of the basis states $|00\rangle\,, |01\rangle\,, |10\rangle\,, |11\rangle$, respectively. The state $|\psi\rangle$ can be written as a tensor product of two single-qubit states if and only if:

$$c_0 c_3 = c_1 c_2 \tag{35}$$

Equivalently, if we reshape the coefficients into a matrix

$$C = \begin{pmatrix} c_0 & c_1 \\ c_2 & c_3 \end{pmatrix}$$

The state is separable if and only if $\det(C) = 0$. More generally, for n-qubit states, there is **no single simple scalar condition** that characterizes separability. A state is separable **if and only if it can be written as a tensor product of n single-qubit states**, which is a more complex condition to verify. So, the 2-qubit case is simpler to analyze, but for larger systems, determining whether a state is separable or entangled can be more challenging.

Let's make this more concrete with two examples:

✔ Let's start from a 4D vector:

$$|\psi\rangle = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = \frac{1}{2}\left(|00\rangle + |01\rangle + |10\rangle + |11\rangle\right)$$

We can check if it is a tensor product by verifying the condition in Equation 35:

$$c_0c_3 = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}, \quad c_1c_2 = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

Since $c_0c_3 = c_1c_2$, the state is a tensor product and thus separable.

Now, we can find the two single-qubit states whose tensor product gives us $|\psi\rangle$. We try:

$$|v_A\rangle = \begin{pmatrix} a_0 \\ a_1 \end{pmatrix}, \quad |v_B\rangle = \begin{pmatrix} b_0 \\ b_1 \end{pmatrix}$$

We want:

$$(a_0b_0, a_0b_1, a_1b_0, a_1b_1) = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right) \quad \Rightarrow \quad \begin{cases} a_0b_0 = \frac{1}{2} \\ a_0b_1 = \frac{1}{2} \\ a_1b_0 = \frac{1}{2} \\ a_1b_1 = \frac{1}{2} \end{cases}$$

Solving this system of equations, we find:

$$a_0 = b_0 = \frac{1}{\sqrt{2}}, \quad a_1 = b_1 = \frac{1}{\sqrt{2}}$$

And this works because:

$$\begin{cases} a_0b_0 = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} \\ a_0b_1 = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} \\ a_1b_0 = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} \\ a_1b_1 = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} \end{cases}$$

Therefore:

$$|\psi\rangle = \left[\frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right)\right] \otimes \left[\frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right)\right]$$

The global 2-qubit state can be written as **two independent qubit states**, confirming that it is separable.

$$|\psi\rangle = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = \left[\frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right)\right] \otimes \left[\frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right)\right]$$

✘ Let's start from a different 4D vector:

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right)$$

We check the condition for separability:

$$c_0c_3 = \frac{1}{2}, \quad c_1c_2 = 0$$

Since $c_0c_3 \neq c_1c_2$, the state is not a tensor product and thus is **not separable**. This state is an example of an **entangled state**, where the two qubits are correlated in such a way that we cannot describe them independently. We will see more about entanglement in the next section, but for now, the key takeaway is that not all multi-qubit states can be factored into tensor products of single-qubit states, and this is a fundamental aspect of quantum mechanics that leads to phenomena like entanglement.

In summary, the tensor product generates only a subset of the full 4-dimensional space of two-qubit states. Separable states have a special multiplicative structure and lie on a lower-dimensional manifold within this space. Most states in the full space do not satisfy this structure and are therefore entangled.

4.1.2 Building a Two-Qubit System

In the previous sections, we have seen how to represent the state of a single qubit using a two-dimensional complex vector. Now, we want to extend this representation to a system of two qubits. To do this, we will use the concept of the tensor product seen in the previous section.

We start from two independent qubits:

$$|v_A\rangle = a_0 |0\rangle + a_1 |1\rangle \quad \text{and} \quad |v_B\rangle = b_0 |0\rangle + b_1 |1\rangle$$

And our **goal is to find a state that encodes all joint possibilities** of the two qubits:

- Both qubits are in state $|0\rangle$: $|00\rangle$
- The first qubit is in state $|0\rangle$ and the second in state $|1\rangle$: $|01\rangle$
- The first qubit is in state $|1\rangle$ and the second in state $|0\rangle$: $|10\rangle$
- Both qubits are in state $|1\rangle$: $|11\rangle$

To represent these joint possibilities, we will use the tensor product of the two qubits' states:

$$|v_A v_B\rangle = |v_A\rangle \otimes |v_B\rangle$$

Expanding this expression, we get:

$$\begin{aligned} |v_A v_B\rangle &= (a_0 |0\rangle + a_1 |1\rangle) \otimes (b_0 |0\rangle + b_1 |1\rangle) \\ &= a_0 b_0 |00\rangle + a_0 b_1 |01\rangle + a_1 b_0 |10\rangle + a_1 b_1 |11\rangle \end{aligned} \tag{36}$$

This state $|v_A v_B\rangle$ is a linear combination of the four possible joint states of the two qubits. Each term in the expansion corresponds to one of the joint possibilities, and the coefficients $a_0 b_0$, $a_0 b_1$, $a_1 b_0$, and $a_1 b_1$ represent the probability amplitudes of each joint state.

We are no longer describing two systems separately, we are describing **one system with 4 possible outcomes**.

General Two-Qubit State

In Equation 36 we built a two-qubit state starting from a **very specific structure**, where the coefficients of the state were products of the coefficients of the individual qubits (i.e., $a_i b_j$). However, we can also consider a more **general two-qubit state**, where the coefficients are not necessarily products of the individual qubits' coefficients:

$$|v\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle \tag{37}$$

Where c_0, c_1, c_2, and c_3 are **arbitrary complex numbers** that represent the probability amplitudes of each joint state.

Obviously, this general two-qubit state **must satisfy the normalization condition**:

$$|c_0|^2 + |c_1|^2 + |c_2|^2 + |c_3|^2 = 1 \tag{38}$$

This means that the sum of the probabilities of all joint states must equal 1, ensuring that the state is properly normalized.

We can also generalize deeper to n qubits, where the state of the **system is a linear combination** of 2^n **basis states**, and the **coefficients** are arbitrary **complex numbers** that satisfy the normalization condition. Mathematically, we can express this as:

$$|v\rangle = \sum_{k=0}^{2^n-1} c_k |k\rangle \tag{39}$$

Where $|k\rangle$ represents the basis states of the n-qubit system, and c_k are the corresponding probability amplitudes. As we have already discussed at page 80, this general form illustrates the exponential growth of the state space as the number of qubits increases. However, we can write our one- and two-qubit states as special cases from this general form:

- For $n = 1$ qubit, we have $2^1 = 2$ basis states: $|0\rangle$ and $|1\rangle$, and the state can be written as:
$$|v\rangle = c_0 |0\rangle + c_1 |1\rangle$$
- For $n = 2$ qubits, we have $2^2 = 4$ basis states: $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$, and the state can be written as:
$$|v\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle$$

In summary, a general two-qubit state is a normalized superposition (i.e., the *coefficients satisfy the normalization condition*) of the four basis states (i.e., $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$). Unlike tensor product states, its coefficients are arbitrary and *may not* factorize, leading to entanglement and non-classical correlations between the qubits (like the state in Example at page 84).

4.1.3 Separable vs Entangled States

Let the general state of a two-qubit system be (as defined in Equation 37, page 86):

$$|v\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle$$

The question is: *can we interpret this as two independent qubits?* The answer is **not always**. As we saw on page 82, the answer depends on whether the states are separable or entangled.

Separable States

A state is **separable if it can be written as a tensor product of two single-qubit states**. In other words, there exist single-qubit states $|v_A\rangle$ and $|v_B\rangle$ such that:

$$|v\rangle = |v_A\rangle \otimes |v_B\rangle$$

So the system can be split into one state for qubit A and one state for qubit B, and the global state is just the combination of these two states. As consequence, the coefficients factorize as:

$$c_0 = a_0 b_0, \quad c_1 = a_0 b_1, \quad c_2 = a_1 b_0, \quad c_3 = a_1 b_1$$

Entangled States

A state is **entangled if it cannot be written as a tensor product of two single-qubit states**. In other words, there do not exist single-qubit states $|v_A\rangle$ and $|v_B\rangle$ such that:

$$|v\rangle \neq |v_A\rangle \otimes |v_B\rangle$$

In this case, the state cannot be separated into two independent qubits, and the coefficients do not factorize as above. The system is simply one single entity. For example, the Bell state (we will see in future sections) is an entangled state:

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}} \left(|00\rangle + |11\rangle\right)$$

And the demonstration was written on page 84. In other words, we can only say that the system is in state $|\Phi^+\rangle$, but we cannot say that qubit A is in some state and qubit B is in some state, because the state of the system does not factorize into two independent states.

At this stage, entanglement is introduced as a structural property of quantum states. Its physical meaning and implications become clear only when studying measurement and quantum circuits in later sections.

Difference. With **separable** states, the information is stored in the individual qubits. In contrast, with **entangled** states, the information is stored in the correlations between the qubits.

4.1.4 Why Entanglement Matters

Let's answer to the question "*why entanglement matters*" step by step. First of all, the **complexity of quantum states grows exponentially** with the number of qubits, as we have introduced at page 80. As consequence, to describe the system, we need 2^n complex numbers, where n is the number of qubits. This means that for a system of 50 qubits, we would need 2^{50} complex numbers to describe its state, which is an astronomically large number (approximately 1.13×10^{15}). It could be a **problem for classical computers**, but quantum computers can handle this complexity naturally.

The reason why quantum computers can handle this complexity naturally is not that they explicitly store all these amplitudes as a classical computer would. Rather, the quantum system itself physically evolves in this exponentially large state space.

Entanglement matters because **most quantum states cannot be decomposed into independent single-qubit states**. In fact, the set of separable states, i.e., states that can be written as a tensor product of individual qubit states, is **very small compared to the set of all possible states**. For two qubits, separable states must satisfy:

$$c_0 c_3 = c_1 c_2$$

Which is a strong structural constraint. Therefore, most two-qubit states do not satisfy this condition and are entangled.

But why does this matter? Because **entanglement prevents the system from being compressed into simpler independent components**. The state must be described globally. This global structure is difficult to simulate classically, but quantum mechanics allows us to manipulate it directly through quantum operations. This is one of the reasons why quantum computers can solve certain problems more efficiently than classical computers.

In summary, entanglement is a fundamental feature of quantum mechanics that prevents a multi-qubit system from being decomposed into independent subsystems. As the number of qubits increases, the state space grows exponentially, and entanglement forces a global description of the system. This global structure can be exploited by quantum algorithms to achieve computational advantages over classical approaches.

4.1.5 Exercise: Normalization of Tensor Product States

Given two qubits in the states $|v_A\rangle$ and $|v_B\rangle$ respectively:

$$|v_A\rangle = a_0 |0\rangle + a_1 |1\rangle , \quad |v_B\rangle = b_0 |0\rangle + b_1 |1\rangle$$

And the combined state of the two qubits is given by the tensor product:

$$|v_A v_B\rangle = |v_A\rangle \otimes |v_B\rangle = a_0 b_0 |00\rangle + a_0 b_1 |01\rangle + a_1 b_0 |10\rangle + a_1 b_1 |11\rangle$$

Show that:

$$|a_0 b_0|^2 + |a_0 b_1|^2 + |a_1 b_0|^2 + |a_1 b_1|^2 = 1$$

Proof. To show that the sum of the squares of the coefficients equals 1, we start expanding the expression:

$$|a_0 b_0|^2 + |a_0 b_1|^2 + |a_1 b_0|^2 + |a_1 b_1|^2$$

Using the property:

$$|xy|^2 = |x|^2 |y|^2$$

Applying this property to each term, we get:

$$\begin{aligned} &= |a_0|^2 |b_0|^2 + |a_0|^2 |b_1|^2 + |a_1|^2 |b_0|^2 + |a_1|^2 |b_1|^2 \\ &= |a_0|^2 \left(|b_0|^2 + |b_1|^2\right) + |a_1|^2 \left(|b_0|^2 + |b_1|^2\right) \end{aligned}$$

Since $|v_B\rangle$ is a valid quantum state and must be normalized, we have:

$$|b_0|^2 + |b_1|^2 = 1$$

Substituting this back into our expression, we get:

$$\begin{aligned} &= |a_0|^2 \cdot 1 + |a_1|^2 \cdot 1 \\ &= |a_0|^2 + |a_1|^2 \end{aligned}$$

Since $|v_A\rangle$ is also a valid quantum state and must be normalized, we have:

$$|a_0|^2 + |a_1|^2 = 1$$

Therefore, we conclude that:

$$|a_0 b_0|^2 + |a_0 b_1|^2 + |a_1 b_0|^2 + |a_1 b_1|^2 = 1$$

The combined state $|v_A v_B\rangle$ is also a valid quantum state and must be normalized, which is consistent with our result. QED

This exercise demonstrates that the **tensor product of two normalized quantum states is also normalized**, because the squared amplitudes factorize and the normalization conditions of the individual states ensure that the total probability remains 1.

4.2 Introduction to Multiple Qubit Gates

Until now, we learned that a quantum state is a vector in a complex vector space, and for 2 qubits, the state is a vector in $\mathbb{C}^4$:

$$|v\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle$$

Now the question is: ***how do we change this state?***

The key idea. A **quantum gate** is just a rule that transforms one state into another. For 1 qubit we already know (page 38):

$$|v'\rangle = U |v\rangle$$

Where U is a 2×2 unitary matrix. For example, the Hadamard gate H (page 61) transforms the state $|0\rangle$ into $|+\rangle$:

$$H |0\rangle = |+\rangle$$

What about 2 qubits? For 2 qubits, we need a 4×4 unitary matrix to transform the state:

$$|v'\rangle = U |v\rangle \quad \text{with} \quad |v\rangle = \begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{pmatrix}, \quad U = \begin{pmatrix} u_{00} & u_{01} & u_{02} & u_{03} \\ u_{10} & u_{11} & u_{12} & u_{13} \\ u_{20} & u_{21} & u_{22} & u_{23} \\ u_{30} & u_{31} & u_{32} & u_{33} \end{pmatrix}$$

This means that the gate can change **all amplitudes at once**, not just one at a time as in the single-qubit case. This is the important difference between single and multi-qubit gates: with 1 qubit, we modify only 2 amplitudes, but with 2 qubits, we can modify all 4 amplitudes **simultaneously**. And now, the transformation can mix amplitudes, create correlations, and even generate entanglement between the qubits.

In simple terms, a multi-qubit gate is just a transformation of a larger state vector. Nothing more, nothing less. However, this simple idea has important implications: it allows us to manipulate the global state of a quantum system, including correlations between qubits, which can lead to computational advantages over classical approaches.

4.3 Tensor Product of Quantum Gates

In a quantum system composed of multiple qubits, it is common to **apply gates independently and simultaneously to different qubits**. This operations is known as parallel execution, and it is described formally using the **Tensor Product of Quantum Gates**.

Motivation

Suppose we have two qubits in individual states:

$$|v_A\rangle = a_0 |0\rangle + a_1 |1\rangle \qquad |v_B\rangle = b_0 |0\rangle + b_1 |1\rangle$$

The **joint state** of the system is represented using the tensor product (page **??**):

$$|v_{AB}\rangle = |v_A\rangle \otimes |v_B\rangle$$

Now assume we wish to apply:

- A gate A to the first qubit $|v_A\rangle$
- A gate B to the second qubit $|v_B\rangle$

How do we describe this *combined operation*?

Formal Definition

The combined transformation on the system is represented by the **tensor product of the operators**:

$$(A \otimes B)(v_A \otimes v_B)$$

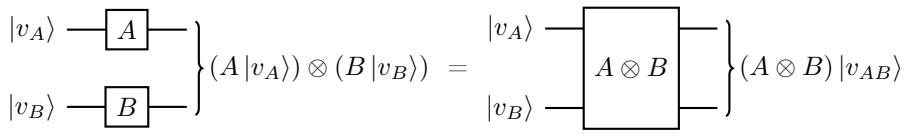
This is **not the same as applying** A and B **separately** and then multiplying the results. It's a structured, mathematically-defined operation where:

- A acts only on the first qubit
- B acts only on the second qubit

The overall effect on the joint state v_{AB} is captured by applying the composite operator $A \otimes B$.

Circuit Interpretation

In a quantum circuit diagram, each horizontal line represent a qubit, and gates are applied along these lines. The parallel application of gates is captured mathematically by $A \otimes B$, and visually by **applying gates side-by-side in the circuit**.



Applying a Gate to One Qubit

Let's say we have two qubits:

$$|v_{AB}\rangle = v_A \otimes v_B$$

We want to apply:

- No operation to qubit A (i.e., we want it to stay as it is)
- Gate B to qubit B

But since we're working in the **joint system**, we need to describe the total gate as acting on the full state vector in $\mathbb{C}^4$. We cannot apply gate B directly on the 2nd qubit in isolation.

To formalize "*do nothing*" to qubit A, we use the **Identity Gate**:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \tag{40}$$

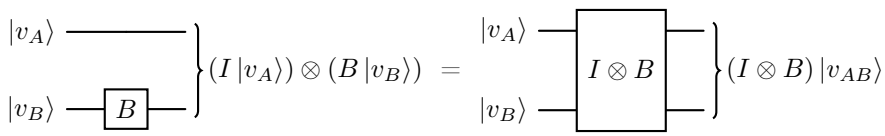
Then, the correct total operation on the two-qubit state is:

$$(I \otimes B)\,(v_A \otimes v_B)$$

This notation ensures that:

- I acts on qubit A and **leaves it unchanged**
- B is applied to qubit B
- The full operator still acts on the **tensor product state space** $\mathbb{C}^4$

In the circuit diagram, this is shown as:



Even if we don't draw the identity gate, it is **implicitly applied** to all untouched qubits, because we are still working in the joint space.

Hadamard Transform on Multiple Qubits

In Section 3.3.4 page 61, we saw how to apply the Hadamard operation to a single qubit system. Now that we are on multiple qubit systems, we want to understand how to use the Hadamard operation.

The **Hadamard Gate (H)** creates superposition states. When applied to each qubit in a system, it **transforms a basis state into a uniform superposition of all possible states**. To apply the Hadamard gate to **two qubits in parallel**, we compute the **tensor product of two Hadamard matrices**:

$$H^{\otimes 2} = H \otimes H$$

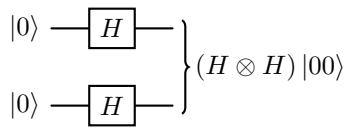
This results in a 4×4 matrix:

$$H^{\otimes 2} = H \otimes H = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

For example, if we want to apply the Hadamard gate to the state $|00\rangle$, mathematically we have:

$$H^{\otimes 2} |00\rangle = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \frac{1}{2} \left(|00\rangle + |01\rangle + |10\rangle + |11\rangle \right)$$

This is a **uniform superposition over all possible two-qubit basis states**. From a circuit point of view:



The **Hadamard operation can be generalized** to n qubits:

$$H^{\otimes n} |0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{i=0}^{2^n - 1} |i\rangle \tag{41}$$

This creates a uniform quantum state across all 2^n possibilities. Something **impossible in classical computing** without explicitly listing all combinations.

4.4 Controlled NOT (CNOT) Gate

The **Controlled-NOT (CNOT)** gate is one of the most fundamental and widely used **two-qubit gates** in quantum computing. It serves as a canonical example of a **quantum gate that cannot be decomposed into single-qubit operations**, and it plays a central role in the generation of entanglement, error correction, and universal gate constructions.

How does it work?

The CNOT gate operates on two qubits:

- The **control qubit** $|c\rangle$
- The **target qubit** $|t\rangle$

Its defining action is **conditional flipping of the target qubit**. If the control qubit is in the state:

- $|1\rangle$, the CNOT gate **applies the Pauli-X (NOT) gate** (page 47) to the target (so the target qubit flips and 0 becomes 1 and 1 becomes 0).
- $|0\rangle$, the **target is left unchanged**.

This behavior can be expressed algebraically as:

$$\text{CNOT}\,|c\rangle\,|t\rangle = |c\rangle\,|t \oplus c\rangle \tag{42}$$

Where:

- Qubit Input: two-qubit basis state $|c\rangle|t\rangle$ ($|c\rangle\,|t\rangle = |c\,t\rangle$).
- CNOT operator: if $c = 0$, leave t unchanged, otherwise, flip t.
- XOR $\oplus$ symbol: denotes addition modulo 2.

  $$t \oplus c = \begin{cases} 0 & \text{if } t = c \\ 1 & \text{if } t \neq c \end{cases}$$

  It flips t if and only if $c = 1$. The **Addition modulo 2**, often denoted as $\oplus$, is a simple binary operation where the result is the remainder after dividing the sum of two bits by 2. It's also known as the **exclusive OR (XOR)** operation.

a	b	$a \oplus b$
0	0	0
0	1	1
1	0	1
1	1	0

  So, $t \oplus c$ means: "flip the target qubit t if and only if the control qubit c is 1".

- The control qubit $|c\rangle$ stays the same.
- The target qubit $|t\rangle$ becomes $|t \oplus c\rangle$ (flipped if $c = 1$).

Matrix Representation

The matrix representation of the CNOT gate is:

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \tag{43}$$

It operates on **two-qubit states** ordered as:

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$$

This matrix is clearly **unitary**, preserving the norm of quantum states, and it **interchanges the basis** vectors $|10\rangle$ and $|11\rangle$ while **leaving** $|00\rangle$ and $|01\rangle$ **unchanged**, consistent with its definition.

For example, let a general two-qubit quantum state written as a linear combination of basis states:

$$|\psi\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle$$

Where:

$$c_0, c_1, c_2, c_3 \in \mathbb{C} \quad \text{and} \quad \sum |c_i|^2 = 1$$

Now apply the CNOT gate:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} c_0 \\ c_1 \\ c_3 \\ c_2 \end{bmatrix}$$

The amplitudes of the $|10\rangle$ and $|11\rangle$ components have been swapped. This corresponds to flipping the target qubit only when the control is $|1\rangle$:

$$|\psi\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |11\rangle + c_3 |10\rangle$$

Who is the Control Qubit?

We have discussed that the CNOT gate operates on two qubits: control $|c\rangle$ and target $|t\rangle$. In the operations, *how can we understand which is the control qubit or the target qubit?* By **convention**, the **control qubit is the most significant qubit**, i.e., the **leftmost one** in the tensor product $|c\,t\rangle$. So:

- $|00\rangle$: control $= 0$, target $= 0$
- $|01\rangle$: control $= 0$, target $= 1$
- $|10\rangle$: control $= 1$, target $= 0$
- $|11\rangle$: control $= 1$, target $= 1$

Thus, in a state like:

$$|\psi\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle$$

The **first qubit is the control**, and the **second is the target**.

Generate Entanglement

An important aspect of the CNOT gate is its **ability to generate entanglement**. For instance, if the control qubit is prepared in a superposition state using a Hadamard gate H, such that the input to the CNOT is:

$$(H \otimes I) |00\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle)$$

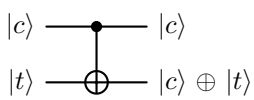
Then applying the **CNOT produces the entangled Bell state**:

$$\text{CNOT} \left(\frac{1}{\sqrt{2}} (|00\rangle + |10\rangle) \right) = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

This outcome exemplifies how **CNOT combined with single-qubit gates can prepare maximally entangled states**, a prerequisite for many quantum protocols including teleportation and superdense coding.

Visual representation of a CNOT

Finally, in quantum circuit diagrams, the CNOT is represented with a solid dot on the control qubit and a $\oplus$ symbol on the target qubit, connected by a vertical line. This visual notation helps emphasize the logical dependence between the two qubits.



4.5 Generic Controlled Gate

After introducing the CNOT gate as controlled-X operation, we generalize the idea to any unitary gate. In quantum computing, we often want to **apply a certain gate U to a target qubit v_B**, but **only if the control qubit v_A is in state $|1\rangle$**. This leads us to the concept of a generic controlled gate, usually denoted CU or C_U.

The **Generic Controlled Gate** has the matrix form:

$$CU = C_U = \begin{bmatrix} I & 0 \\ 0 & U \end{bmatrix} \tag{44}$$

This matrix acts on a two-qubit system, where:

- I is the 2×2 **identity** matrix
- U is the *any* **single-qubit unitary gate** (Pauli-X, Z, Y, ...)
- The **upper block** (top-left 2×2 block) acts when the **control qubit** is $|0\rangle$
- The **lower block** (bottom-right 2×2 block) applies U when the **control qubit** is $|1\rangle$

This form is **not separable**, so the operator C_U cannot be written as a tensor product of two individual single-qubit gates: $CU \neq A \otimes B$. The reason is simple:

- A **tensor product** like $I \otimes U$ **always** applies U to the second qubit **regardless** of the state of the first qubit.
- But C_U applies U to the second qubit **only if the first qubit is** $|1\rangle$.

So C_U is a **conditional operation** that links the behavior of one qubit to the value of the other.

How it works

Suppose we have two qubits:

$$v_A = a_0 |0\rangle + a_1 |1\rangle \qquad v_B = b_0 |0\rangle + b_1 |1\rangle$$

Their joint state is:

$$v_{AB} = v_A \otimes v_B = \begin{bmatrix} a_0 b_0 \\ a_0 b_1 \\ a_1 b_0 \\ a_1 b_1 \end{bmatrix}$$

Then the result of applying the generic controlled gate is:

$$C_U (v_A \otimes v_B) = \begin{bmatrix} I & 0 \\ 0 & U \end{bmatrix} \begin{bmatrix} a_0 b_0 \\ a_0 b_1 \\ a_1 b_0 \\ a_1 b_1 \end{bmatrix}$$

This matrix selectively applies U to the last two components (i.e., when $v_A = 1$), leaving the first two unchanged.

1. **Control is $|0\rangle$**. If $v_A = |0\rangle$, then:

   - $a_0 = 1$
   - $a_1 = 0$

   Consequently:

$$v_{AB} = \begin{bmatrix} a_0 b_0 \\ a_0 b_1 \\ a_1 b_0 \\ a_1 b_1 \end{bmatrix} = \begin{bmatrix} 1 \cdot b_0 \\ 1 \cdot b_1 \\ 0 \cdot b_0 \\ 0 \cdot b_1 \end{bmatrix} = \begin{bmatrix} b_0 \\ b_1 \\ 0 \\ 0 \end{bmatrix}$$

   Using the generic controlled gate, there is **no change**:

$$C_U\left(v_{AB}\right) = \begin{bmatrix} I \cdot \begin{matrix} b_0 \\ b_1 \end{matrix} \\ U \cdot \begin{matrix} 0 \\ 0 \end{matrix} \end{bmatrix}$$

   Because the upper block is multiplied by the identity matrix. So the generic controlled gate **does nothing** when the control qubit is in state $|0\rangle$.

2. **Control is $|1\rangle$**. If $v_A = |1\rangle$, then:

   - $a_0 = 0$
   - $a_1 = 1$

   Now:

$$v_{AB} = \begin{bmatrix} a_0 b_0 \\ a_0 b_1 \\ a_1 b_0 \\ a_1 b_1 \end{bmatrix} = \begin{bmatrix} 0 \cdot b_0 \\ 0 \cdot b_1 \\ 1 \cdot b_0 \\ 1 \cdot b_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ b_0 \\ b_1 \end{bmatrix}$$

   Using the generic controlled gate:

$$C_U\left(v_{AB}\right) = \begin{bmatrix} I \cdot \begin{matrix} 0 \\ 0 \end{matrix} \\ U \cdot \begin{matrix} b_0 \\ b_1 \end{matrix} \end{bmatrix}$$

   So the gate U is **only applied to the target qubit when the control qubit is** $|1\rangle$.

4.6 SWAP Gate

The **SWAP gate** is a two-qubit gate that **exchanges the states of the two qubits**. That is, if qubit A is in state v_A and qubit B in state v_B, then after applying SWAP, their roles are reversed.

Matrix Representation

The SWAP gate is defined by the following 4×4 unitary matrix:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{45}$$

This matrix operates on the basis states in the usual order:

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$$

And swaps the coefficients of $|01\rangle$ and $|10\rangle$, leaving $|00\rangle$ and $|11\rangle$ unchanged.

How it works

Suppose we have two qubits in superposition:

$$v_A = a_0 |0\rangle + a_1 |1\rangle \qquad v_B = b_0 |0\rangle + b_1 |1\rangle$$

Their joint state is:

$$v_{AB} = v_A \otimes v_B = a_0 b_0 |00\rangle + a_0 b_1 |01\rangle + a_1 b_0 |10\rangle + a_1 b_1 |11\rangle$$

After applying the SWAP gate:

$$\text{SWAP}(v_{AB}) = a_0 b_0 |00\rangle + a_1 b_0 |01\rangle + a_0 b_1 |10\rangle + a_1 b_1 |11\rangle$$

But this result can be obtained also doing $v_B \otimes v_A$:

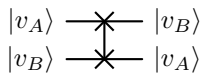
$$v_{BA} = b_0 a_0 |00\rangle + b_0 a_1 |01\rangle + b_1 a_0 |10\rangle + b_1 a_1 |11\rangle$$

Note that the coefficients a_i and b_j are scalar: $a_i \cdot b_j = b_j \cdot a_i$. So the SWAP gate effectively implements:

$$v_A \otimes v_B \xrightarrow{\text{SWAP gate}} v_B \otimes v_A \tag{46}$$

Circuit Representation

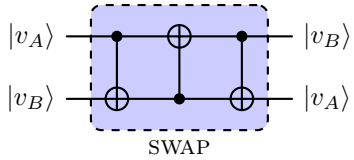
In circuit diagrams, the SWAP gate is often draw using **crossed lines with $\times$ symbols**:



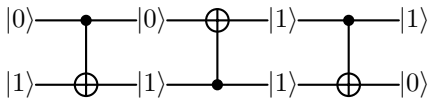
★ SWAP as Three CNOTs

An important implementation detail is that **SWAP can be decomposed into three CNOT gates** (page 95). This is crucial **because SWAP is not a native gate on most quantum hardware**, but CNOT often is.

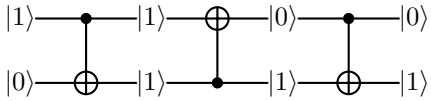
The decomposition is represented graphically as:



1. First CNOT:
   - Control qubit: v_A
   - Target qubit: v_B
2. Second CNOT:
   - Control qubit: v_B
   - Target qubit: v_A
3. Third CNOT:
   - Control qubit: v_A
   - Target qubit: v_B



And:



4.7 Toffoli Gate (CCNOT)

The **Toffoli Gate**, or **CCNOT (Controlled-Controlled-NOT)**, is a **three-qubit gate** where **two qubits act as controls** and **one as the target**. It generalizes the behavior of the CNOT gate by requiring **both control qubits** to be in the state $|1\rangle$ before the NOT operation is applied to the target.

Matrix Representation

The matrix form of the Toffoli gate is an 8×8 identity matrix with only two elements swapped:

$$\text{CCNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

This matrix swpas the $|110\rangle \leftrightarrow |111\rangle$ entries. In other words, all basis states remain unchanged except:

- $|110\rangle$ becomes $|111\rangle$
- $|111\rangle$ becomes $|110\rangle$

Gate Behavior

The CCNOT gate can be described as follows:

- **Inputs**: v_0, v_1 as *control qubits*, and v_2 as *target qubit*.
- **Output**:
$$v_2 \to v_2 \oplus (v_0 \wedge v_1)$$

This means:

- If v_0 logically combined with v_1 returns true, then flip v_2. In other words, flip if and only if v_0 and v_1 are 1.
- Otherwise, leave v_2 unchanged.

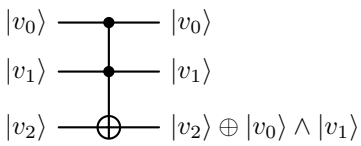
This behavior is **nonlinear** in the classic sense, and it's what makes Toffoli especially powerful: it can **simulate universal classical logic** within a quantum system. So any logic circuit that we can build using AND, OR and NOT gates, can also be implemented using only Toffoli gates.

Circuit Notation

In quantum circuit diagrams, the CCNOT gate is drawn with:

- Two control dots on the top two lines (for v_0, v_1)
- A $\oplus$ symbol on the third line (for the target qubit v_2)
- Vertical lines connecting the three

This shows the conditional behavior clearly: the **NOT is triggered only when both controls are active**.



4.8 Foundations of Universal Quantum Circuits

To implement arbitrary quantum computations, we must answer two fundamental questions:

1. **Which gates are sufficient** to build any quantum circuit?
2. **How do we mathematically model** a circuit built from those gates?

This section addresses both: it first introduces the concept of **universal gate sets**, and then explains how to formally represent quantum circuits as **matrices** acting on the full system state.

Universal Quantum Gate Sets

Quantum computation occurs within the framework of unitary transformations on quantum states. However, since the space of all unitary operations is infinite (continuous), we cannot construct every possible unitary matrix exactly with a finite number of gates. Instead, we aim for **approximate universality**: the ability to approximate any unitary operation to arbitrary precision using a small set of elementary gates.

A **Universal Quantum Gate Set** is a finite collection of quantum gates from which **any unitary operation** on any number of qubits can be approximated arbitrarily well.

Example 2

One of the most commonly used universal sets includes:

$$\{H, T, \text{CNOT}\}$$

- H: Hadamard gate, creates superposition
- T: $\frac{\pi}{8}$ phase gate, a non-Clifford gate necessary for universality
- CNOT: a two-qubit entangling gate

With these gates alone, it is possible to approximate any quantum circuit. This is backed by the Solovay-Kitaev, which ensures efficient approximation of unitaries using such a discrete set.

The set of Clifford gates alone is **not universal**. To achieve universality, we must include **at least one non-Clifford gate**, like T gate. Without it, the resulting circuits can be efficiently simulated classically, and therefore cannot exhibit true quantum advantage.

Modelling Quantum Circuits as Matrices

Quantum circuits are not only visual tools; they correspond to precise mathematical transformations. Every complete circuit defines a **unitary matrix** acting on the system's joint state vector. Understanding how to construct this matrix from a sequence of gates is crucial for analysis and simulation.

When building the matrix of a circuit composed of multiple gates and qubits, we follow three formal rules:

1. **Tensor product across qubits**. Gates applied simultaneously on different qubits are composed using the **tensor product**. For example:

$$A \text{ on qubit } 1, \quad B \text{ on qubit } 2 \Rightarrow A \otimes B$$

2. **Matrix product along time (sequential composition)**. Gates applied **in sequence** (over time) are composed via **matrix multiplication**, applied **from right to left**. This reflects the order of function composition.

3. **Identity padding for untouched qubits**. If a gate acts only on some of the qubits, we insert the **identity matrix** I for those that are not affected, to preserve the tensor product structure.

4.9 Entanglement

In a multi-qubit quantum system, **Entanglement** refers to the **phenomenon in which the quantum state of each qubit cannot be described independently of the state of the orders**. Entangled states are **non-separable**, meaning they cannot be expressed as the tensor product of individual qubit states.

- **The state of a n-qubit system is described by 2^n complex amplitudes.**

  Why? Because a general quantum state is a **superposition of all basis states**. For n qubits, there are 2^n possible basis states (e.g., $|000\rangle$, $|001\rangle$, ..., $|111\rangle$). So a general state is:

  $$|\psi\rangle = c_0 |0\ldots0\rangle + c_1 |0\ldots1\rangle + \cdots + c_{2^n-1} |1\ldots1\rangle$$

  Where:

  - $c_0, c_1, \ldots, c_{2^n-1} \in \mathbb{C}$.
  - So we need 2^n complex numbers $= 2 \cdot 2^n = 2^{n+1}$ real numbers.

- **Accounting for normalization and the irrelevance of global phase, it takes $2^{n+1} - 2$ real numbers to fully specify the state.**

  If we want to **completely describe the quantum state of an n-qubit system**, so that we can simulate it, reconstruct it, or predict its evolution, **we need to know $2^{n+1} - 2$ real numbers**.

  In other words, if someone gives us:

  - A list of $2^{n+1} - 2$ real values
  - And tells us "these specify the quantum state"

  Then:

  1. We can **reconstruct the full state vector**
  2. We know everything we need to calculate outcomes of **any measurable**
  3. We can simulate unitary evolution, or compute entanglement, etc.

Example 3: $n = 1$ qubit

The state is:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle \quad \text{with } \alpha, \beta \in \mathbb{C}$$

That's 4 real numbers (2 complex), but minus:

- 1 for normalization: $|\alpha|^2 + |\beta|^2 = 1$
- 1 for global phase: $e^{i\theta}$ is physically irrelevant

So, to fully describe a single qubit, we need:

$$2^{1+1} - 2 = 2 \text{ real numbers}$$

And this make sense, because we can describe any pure qubit with two real angles θ, ϕ on the block sphere (page **??**).

Why 2^{n+1}?

- 2^n complex amplitudes for an n-qubit system (because there are 2^n basis states).
- Each complex number is made up of 2 real numbers: the real part and the imaginary part. So we have to add one to n.

Why do we subtract 2?

- **Normalization**. **Quantum states must always be normalized**:

$$\sum_{i=0}^{2^n-1} |c_i|^2 = 1$$

  This is **one constraint** on the set of amplitudes. Since it's a real-valuated equation, it removes 1 real degree of freedom from the total.
- **Global Phase**. In quantum mechanics, multiplying a quantum state by a global phase $e^{i\theta}$ doesn't change anything physically measurable:

$$|\psi\rangle \quad \text{and} \quad e^{i\theta} |\psi\rangle$$

  Are **physically indistinguishable**, they lead to the **same measurement outcomes**. So we treat two such states as equivalent. That means we're overcounting one extra complex degree of freedom, which corresponds to the global phase. This removes another real parameter (the angle θ).

- **In contrast, n independent (separable, then not entangled) qubits would only require $2n$ real parameters.**

  What about independent qubits? If the qubits are not entangled, each qubit can be written separately as:

$$|\psi_i\rangle = \alpha_i |0\rangle + \beta_i |1\rangle$$

  Each qubit needs:

  - $\alpha_i, \beta_i \in \mathbb{C}$
  - But again, we subtract 2 real numbers (normalization and global phase) for each qubit.

  So one single qubit state needs **2 real parameters**.

- **Since $2^n \gg 2n$, most quantum states are entangled.**

  What does this mean? If a state can be written as:

  $$v_1 \otimes v_2 \otimes \cdots \otimes v_n$$

  Then it only uses $2n$ real numbers. But a general n-qubit state needs $2^{n+1} - 2 \gg 2n$ real numbers. So: **almost all quantum states cannot be written as tensor products of single-qubit states**. These are the entangled states.

  How is this possible? The **full space** of quantum states is **exponentially larger** $(2^{n+1} - 2)$ than the subspace of separable states $(2n)$. We can think of the space of all quantum states as a huge-dimensional sphere of radius 1 (because of normalization).

  - The set of separable states is a tiny curved surface embedded inside that space.
  - It's so tiny that if we pick a random point on the full sphere (a random quantum state), we will almost surely land outside the separable surface.

  The probability of randomly choosing a separable state from the set of all quantum states is essentially **zero**.

  Note that this is not a mathematical trick! This is **experimentally observable**. If we:

  - Generate random quantum states (e.g. using random circuits)
  - Or simulate the uniform distribution over the Hilbert space

  We'll find that:

  - With high probability, the state cannot be written as a product of individual qubit states.
  - In fact, the expected entanglement entropy of a randomly chosen pure state is very close to maximal.

Example 4: Bell State

One of the canonical examples of an entangled state is the **Bell state**:

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right) \tag{47}$$

Suppose we try to write this as product of two qubits:

- Let $v_A = a_0 |0\rangle + a_1 |1\rangle$
- Let $v_A = b_0 |0\rangle + b_1 |1\rangle$

Their tensor product becomes:

$$v_A \otimes v_B = a_0 b_0 |00\rangle + a_0 b_1 |01\rangle + a_1 b_0 |10\rangle + a_1 b_1 |11\rangle$$

To match the Bell state, we would require:

$$a_0 b_0 = \frac{1}{\sqrt{2}}, \quad a_1 b_1 = \frac{1}{\sqrt{2}}, \quad a_0 b_1 = 0, \quad a_1 b_0 = 0$$

But if $a_0 b_1 = 0$, then at least one of a_0 or b_1 must be zero. Similarly, if $a_1 b_0 = 0$, one of a_1 or b_0 must be zero. This leads to a contradiction, because we also need $a_0 b_0 \neq 0$ and $a_1 b_1 \neq 0$. Therefore, no such a_0, a_1, b_0, b_1 exist.
Conclusion: the Bell state cannot be written as a tensor product of two individual qubit states, it is entangled.

The Nature of Entanglement

Entanglement is a **structural property** of the quantum state, **not dependent on the basis**. This **contrasts with superposition**, which is basis-dependent.

- Whether a state is entangled depends on the decomposition of the system.
- A state can be entangled with respect to one partitioning of subsystems, and separable under another.
- In most contexts, when we say **a state is entangled**, we refer to its **entanglement with respect to the individual qubits** (standard tensor product structure).

So we can change the basis of the individual qubits (rotate them, use Hadamard basis), but we will **not** be able to write Bell state state as $|\psi_A\rangle \otimes |\psi_B\rangle$. No matter what basis we use, we will **not** be able to "break" the entanglement. Entanglement is about whether the state **can be written as a product of parts**, this is a property of the **whole structure**, not of how we write it.

Entangling Gates

A gate is said to be **entangling** if it is **capable of producing entangled states from separable inputs**. Not all two-qubit gates are entangling.

- A **generic 2-qubit gate** is a 4×4 unitary matrix, with 16 complex entries (minus constraint from unitarity).
- Two single-qubit gates acting independently on two qubits use only $2 \times 4 = 8$ complex parameters.

Thus, **most two-qubit gates are not tensor products** of single-qubit gates. If a gate cannot be written as $U_1 \otimes U_2$, it is **non-separable** and **can generate entanglement**.

Example 5: CNOT as an Entangling Gate

The CNOT gate can transform separable states into entangled states, for example:

$$\text{CNOT}\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |0\rangle\right) = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \left|\Phi^+\right\rangle \quad \text{(Bell state)}$$

This demonstrates the key role of CNOT in quantum-algorithms, it **creates entanglement**, which is a resource for quantum advantage.

What are Bell States?

Bell States are **four special two-qubit states**:

$$\begin{aligned} |\Phi^+\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) & |\Phi^-\rangle &= \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \\ |\Psi^+\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) & |\Psi^-\rangle &= \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \end{aligned} \tag{48}$$

That are:

- **Maximally entangled**: measuring one qubit fully determines the other.
- **Orthonormal**: they form a basis of the two-qubit Hilbert space.
- **Used in fundamental protocols** like teleportation, superdense coding, and entanglement-based quantum cryptography.

Bell states represent **all possible maximally entangles superpositions of two-qubit basis states**.

Why are they important?

Bell states are not just theoretical constructs, they are:

- **Fundamental to quantum computing**: often used as building blocks for more complex states.
- **Hard to simulate classically**: their entanglement implies that classical representations cannot compress them efficiently.
- **Maximally entangled**: each individual qubit, when observed in isolation, appears **completely random**, even though the total state is pure and deterministic.

When a Bell state is measured on one qubit, the outcome of the second qubit is **fully correlated**, but until that measurement happens, each qubit looks like it is in a **completely mixed state** (i.e. maximum uncertainty). This **contrast with unentangled (separable)** states, where each qubit can be described independently and has less uncertainty when observed locally.

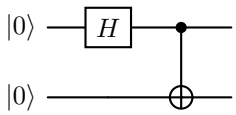
How to create Bell states

Bell states can be created using a **simple 2-gate quantum circuit**:

1. Apply a **Hadamard gate to the first qubit**:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

2. Apply a **CNOT gate**, with:
   - Control: qubit 1
   - Target: qubit 2



The final output is:

$$\text{CNOT}\,(H \otimes I)\,(|00\rangle) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \left|\Phi^+\right\rangle$$

So this circuit creates the Φ^+ Bell state starting from $|00\rangle$. Mathematical:

1. Compute $H \otimes I$, i.e., apply Hadamard to the first qubit:

$$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}$$

2. Then apply the CNOT operator to that result:

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

3. Multiply:

$$\text{CNOT} \cdot (H \otimes I)\,|00\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Example 6: Bell States in the Hadamard Basis

In this exercise, we express the Bell state $|\Phi^+\rangle$ in the Hadamard basis instead of the computational basis.

Bell state $|\Phi^+\rangle$ in the Hadamard basis. Given:

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

We aim to re-express this state using the Hadamard basis $\{|+\rangle, |-\rangle\}$, where:

- $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
- $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

1. **Invert the definitions**. To convert from the computational basis to the Hadamrd basis, we invert the above equations to express $|0\rangle$ and $|1\rangle$ in terms of $|+\rangle$ and $|-\rangle$:
   - $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$
   - $|1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$
2. **Substitute into $|00\rangle$ and $|11\rangle$**. We compute the tensor products using the expression above.
   - Compute $|00\rangle$:
   $$\begin{aligned} |00\rangle &= |0\rangle \otimes |0\rangle \\ &= \left(\frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)\right) \otimes \left(\frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)\right) \\ &= \frac{1}{2}(|++\rangle + |+-\rangle + |-+\rangle + |--\rangle) \end{aligned}$$
   - Compute $|11\rangle$:
   $$\begin{aligned} |11\rangle &= |1\rangle \otimes |1\rangle \\ &= \left(\frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)\right) \otimes \left(\frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)\right) \\ &= \frac{1}{2}(|++\rangle - |+-\rangle - |-+\rangle + |--\rangle) \end{aligned}$$
3. **Add the two to get $|\Phi^+\rangle$**. Now add the two results:
   $$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Substituting:

$$\begin{aligned}
|\Phi^+\rangle &= \frac{1}{\sqrt{2}}\left[\frac{1}{2}\left(|{+}{+}\rangle + |{+}{-}\rangle + |{-}{+}\rangle + |{-}{-}\rangle\right)\right. \\
&\qquad \left. + \frac{1}{2}\left(|{+}{+}\rangle - |{+}{-}\rangle - |{-}{+}\rangle + |{-}{-}\rangle\right)\right] \\
&= \frac{1}{\sqrt{2}} \cdot \frac{1}{2}\left(2\,|{+}{+}\rangle + 0\,|{+}{-}\rangle + 0\,|{-}{+}\rangle + 2\,|{-}{-}\rangle\right) \\
&= \frac{1}{\sqrt{2}} \cdot \frac{1}{2}\left(2\,|{+}{+}\rangle + 2\,|{-}{-}\rangle\right) \\
&= \frac{1}{\sqrt{2}}\left(|{+}{+}\rangle + |{-}{-}\rangle\right)
\end{aligned}$$

Thus, in the Hadamard basis, we have:

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}\left(|{+}{+}\rangle + |{-}{-}\rangle\right)$$

Which is a Bell-like state expressed in the Hadamard basis instead of the computational basis. QED

4.10 Measurement in Multi-Qubit Systems

We begin with a general two-qubit state:

$$|v_C\rangle = |v_A v_B\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle$$

This is an entangled or general superposed state over the two-qubit basis. But the question is: ***What happens to the state of the system if we measure only qubit A?***

Decomposition Strategy

The idea is to **separate the global state** into parts conditioned on the state of qubit A:

$$|v\rangle = c_{01} |0\rangle \otimes \frac{c_0 |0\rangle + c_1 |1\rangle}{c_{01}} + c_{23} |1\rangle \otimes \frac{c_2 |0\rangle + c_3 |1\rangle}{c_{23}}$$

Where:

- $c_{01} = \sqrt{|c_0|^2 + |c_1|^2}$
- $c_{23} = \sqrt{|c_2|^2 + |c_3|^2}$

This expression means: the total state is a **superposition of two conditional branches**:

- One where qubit A is $|0\rangle$, and qubit B is in a normalized superposition of $|0\rangle$, $|1\rangle$.
- One where qubit A is $|1\rangle$, and qubit B is in another normalized state.

What happens after a Measurement?

1. Case 1. **Qubit A is measured as $|0\rangle$**:
   - The outcome collapses the state: $c_2 = 0$, $c_3 = 0$
   - The system is no longer a superposition, it's now in:

     $$|v_B\rangle = \frac{c_0 |0\rangle + c_1 |1\rangle}{c_{01}}$$

     Which is a valid normalized single-qubit state (i.e., $|v_B\rangle\langle v_B| = 1$)
2. Case 2. **Qubit A is measured as $|1\rangle$**:
   - Similarly, $c_0 = 0$, $c_1 = 0$
   - New state of B becomes:

     $$|v_B\rangle = \frac{c_2 |0\rangle + c_3 |1\rangle}{c_{23}}$$

Measuring **one qubit** in an entangled system **instantaneously collapses the entire state**, even though **only one qubit** is directly observed.

This feels strange because **measurement is local**, yet the **effect is nonlocal**, it determines the state of the other qubit, even if it's far away! This is a **manifestation of quantum entanglement** and is deeply **linked to** the phenomenon of **quantum nonlocality**.

Example 7: Bell state

Let's apply this idea to the Bell state:

$$|v\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right)$$

This can be rewritten as:

$$|v\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle\right)$$

So we're in the entangled state $|\Phi^+\rangle$.

Now, we measure qubit A. If we get **outcome**:

- $|0\rangle$, the global state collapses to $|00\rangle \Rightarrow$ qubit B is now in state $|0\rangle$.
- $|1\rangle$, the state collapses to $|11\rangle \Rightarrow$ qubit B is in state $|1\rangle$.

4.11 Limits of Quantum Information

4.11.1 No-Cloning Principle

The **No-Cloning Theorem** states that it is **impossible to create an exact copy of an arbitrary unknown quantum state**. This is not a limitation of current technology, it is a fundamental result derived directly from the **linearity of quantum mechanics**.

Why does it matter?

The no-cloning principle is critical because it prevents:

1. **Violations of quantum mechanics**: cloning would break the rules of **unitarity** and **linearity** of quantum evolution.
2. **No-signaling violation** (section 4.11.3 in depth, page 121): if cloning were possible, we could **instantly send information** across large distances, violating causality.
3. **Infinite data compression**: in theory, we could encode an arbitrarily long classical string into a quantum state and clone it infinitely to retrieve the data using quantum tomography.
4. **Quantum error correction becomes harder**: many classical redundancy techniques rely on copying data, which is not possible in quantum mechanics.

Formal Proof

Let's try to define a hypothetical **cloning unitary operator** U. We assume:

$$U\left(|x\rangle \otimes |0\rangle\right) = |x\rangle \otimes |x\rangle$$

This says: *take any qubit* $|x\rangle$ *and a blank* $|0\rangle$*, and get two* $|x\rangle$. Now take two arbitrary qubits $|x\rangle$ and $|y\rangle$, and apply the same gate to both:

$$U\,|x\rangle\,|0\rangle = |x\rangle\,|x\rangle \quad U\,|y\rangle\,|0\rangle = |y\rangle\,|y\rangle$$

Since U is **a unitary operator**, it **preserves inner products** and the following statement <u>should</u> be true:

$$\langle x \otimes 0 | y \otimes 0\rangle = \langle x \otimes x | y \otimes y\rangle$$

Let's compute both sides:

- Left side, before cloning:

$$\begin{aligned}\langle x \otimes 0 | y \otimes 0\rangle &= \langle x|y\rangle \cdot \langle 0|0\rangle \\ &= \langle x|y\rangle \cdot 1 \\ &= \langle x|y\rangle\end{aligned}$$

- Right side, after cloning:

$$\begin{aligned}\langle x \otimes x | y \otimes y \rangle &= \langle x|y\rangle \cdot \langle x|y\rangle \\ &= \langle x|y\rangle^2\end{aligned}$$

So now're claiming:

$$\langle x|y\rangle = \langle x|y\rangle^2$$

Let's analyze this equation:

- Let $\alpha = \langle x|y\rangle \in \mathbb{C}$
- Then $\alpha = \alpha^2 \Rightarrow \alpha - \alpha^2 = 0 \Rightarrow \alpha(\alpha - 1) = 0$

So this is only true if $\alpha = 0$ or $\alpha = 1$:

- The states are **identical**; same direction, same vector:

$$\langle x|y\rangle = 1$$

- The states are **orthogonal**; completely distinguishable:

$$\langle x|y\rangle = 0$$

Unfortunately, for **arbitrary quantum states**, the inner product is **almost always** somewhere between 0 and 1. Only in **special cases** (perfect alignment or perfect orthogonality) is it exactly 0 or 1. That's why we say:

$$\langle x|y\rangle \neq 0, 1 \text{ in general}$$

This is **exactly why the no-cloning proof works**, because it **fails for general** (non-orthogonal) **input states**, which are the rule rather than the exception.

⚠ Apparent Violation via CNOT

Let's explore a common misconception. We apply a CNOT gate with:

- Control qubit $v_A = a_0 |0\rangle + a_1 |1\rangle$
- Target qubit $v_B = |0\rangle$

Naively, one might write:

$$\text{CNOT}(v_A, v_B) = v_A \otimes (v_A \oplus 0) = v_A \otimes v_A$$

It **looks like we cloned** v_A! But this is <u>incorrect</u> because:

- That informal rule $v_A \oplus v_B$ **only works for classical basis states** ($|0\rangle$ or $|1\rangle$).
- When v_A is in a **superposition**, this model breaks.

What really happens is:

1. Initial state:
$$v_A \otimes |0\rangle = a_0 |00\rangle + a_1 |10\rangle$$

2. Apply CNOT (flip second bit if control is 1):
$$\Rightarrow a_0 |00\rangle + a_1 |11\rangle$$

This is **not** a clone! It's an **entangled state**, we cannot factor it as $|\psi\rangle \otimes |\psi\rangle$. Compare with actual cloned state:

$$(a_0 |0\rangle + a_1 |1\rangle) \otimes (a_0 |0\rangle + a_1 |1\rangle) = a_0^2 |00\rangle + a_0 a_1 |01\rangle + a_0 a_1 |10\rangle + a_1^2 |11\rangle$$

They're **clearly different**: cloning would produce 4 terms, but CNOT produces only 2. Thus CNOT **does not** clone a general qubit. It **creates entanglement**, not copies.

4.11.2 No-Deleting Principle

The **No-Deleting Theorem** states that it is **impossible to delete an arbitrary unknown quantum state**. In other words, **there does not exist a unitary operator** U that can take a pair of identical qubits $|x\rangle \otimes |x\rangle$ and produce:

$$U\left(|x\rangle \otimes |x\rangle\right) = |0\rangle \otimes |x\rangle$$

This operation would "delete" the first copy of $|x\rangle$, replacing it with a standard $|0\rangle$, while leaving the second untouched.

Why is deletion a problem?

If we could delete an arbitrary state, we could:

- Circumvent information preservation in quantum systems (which must be **unitary and reversible**).
- Violate **quantum linearity**, much like cloning would.
- Enable undesirable operations that break the **symmetry and reversibility** of quantum evolution.

Formal Proof

Let's assume that such a deleting unitary U exists. We define:

$$U\left(|x\rangle \otimes |x\rangle\right) = |0\rangle \otimes |x\rangle \quad U\left(|y\rangle \otimes |y\rangle\right) = |0\rangle \otimes |y\rangle$$

Take the inner product between both results:

- Left side, *before* deleting:

  $$\langle x| \langle x| \cdot U^\dagger U \cdot |y\rangle |y\rangle$$

  Because $U^\dagger U = I$, and inner products distribute:

  $$= \langle x|y\rangle \cdot \langle x|y\rangle = \langle x|y\rangle^2$$

- Right side, *after* deleting:

  $$\langle x|0\rangle \cdot \langle y|0\rangle = \langle x|y\rangle$$

  So now we're saying:

  $$\langle x|y\rangle^2 = \langle x|y\rangle \quad \Rightarrow \quad \langle x|y\rangle \cdot (\langle x|y\rangle - 1) = 0$$

  So the only possible solutions are:

  $$\langle x|y\rangle = 0 \quad \text{(orthogonal)} \quad \text{or} \quad \langle x|y\rangle = 1 \quad \text{(identical)}$$

But for **general quantum states**, $\langle x|y\rangle \notin \{0, 1\}$, so we've reached a **contradiction**. Therefore, **no such deletion operator** U **can exist for arbitrary stats**.

Physical Meaning

The no-deleting principle complements the **no-cloning principle**:

- We can't **make** copies of unknown quantum states (no-cloning)
- We can't **erase** arbitrary copies either (no-deleting)

These theorems emphasize that **quantum information is conserved**:

- It cannot be copied freely
- It cannot be erased freely
- It can only be **transferred** (e.g., through teleportation or interaction)

4.11.3 No-Signaling Principle

The **No-Signaling Theorem** asserts that: **no information can be transmitted faster than the speed of light, even using entangled quantum systems**.

Although **entanglement leads to instantaneous correlations** between distant qubits, it **does not allow communication** or information transfer without a classical channel.

This principle **preservers causality** and ensures that quantum mechanics remains compatible with **special relativity**.

Explained Proof

To understand the **no-signal principle** in quantum mechanics, consider the classical scenario involving two observers, Alice and Bob, who share an **entangled pair** of qubits. Specifically, they share the Bell state:

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right)$$

Alice has qubit a, Bob has qubit b, and they are located far apart, even at opposite ends of the universe.

What happens when Alice measures? Suppose Alice chooses to **measure her qubit** in the computational basis.

- If she gets outcome $|0\rangle$, the join state **collapses** to $|00\rangle$
- If she gets $|1\rangle$, the state collapses to $|11\rangle$

In both cases, Bob's qubit becomes **perfectly correlated** with Alice's result. **So yes**, Alice's measurement causes the collapse of Bob's qubit; instantly, no matter the distance.

Can Bob detect this? Here's the crucial point: **Bob cannot tell whether Alice has measured her qubit**.

Why? Because from **Bob's perspective**, before doing anything, his own qubit is described by the **reduced density matrix**:

$$\rho_B = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$$

This is a **completely mixed state**. If Bob measures it, he gets:

- 50% chance of $|0\rangle$
- 50% chance of $|1\rangle$

Now suppose **Alice has already measured**, and the state is now $|00\rangle$.

- Bob's qubit is now **definitely** $|0\rangle$

- But Bob doesn't know that, and he has **no way to know**
- The only way for him to learn this is:
  1. **To measure** (and discover the value)
  2. Or **to be told** by Alice via classical communication

Why this preserves No-Signaling. Even though **Alice's measurement affects the state** of Bob's qubit, **it does not change what Bob sees statistically**, unless he receives a message from her. Therefore:

- Alice **cannot use entanglement** to send messages faster than light.
- Bob's observable outcomes are **indistinguishable** whether Alice measured or not.
- **Entanglement creates correlations**, not communication.

Summary of the story

- Entanglement causes **instant collapse of joint state** when one qubit is measured.
- But the **other party cannot detect** this collapse unless they **measure** or are **told**.
- No observable effect on Bob's side = **no information transfer** = no violation of causality.

Why it matters?

The no-signaling principle **guarantees**:

- ✔ **No faster-than-light communication**, preserving consistency with relativity
- ✔ **No "quantum telepathy"**, even in systems that exhibit entanglement
- ✔ Measurement outcomes are **random locally**, even when entanglement exists globally.

References

[1] Cremonesi Paolo. Quantum computing. Slides from the HPC-E master's degree course on Politecnico di Milano, 2024.

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